Exercises and Solutions

Theory of Distributions

Prepared by ...

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Problem

Let X be a non-compact subset of $\mathbb{R}^2$. Prove that there is a continuous unbounded function from X to $\mathbb{R}$.

Background

Recall that by the Heine–Borel theorem, a subset $K \subseteq \mathbb{R}^2$ is compact if and only if it is closed and bounded. Thus, if X is non-compact, either

1. X is unbounded, or
2. X is bounded but not closed.

A function $f : X \to \mathbb{R}$ is unbounded if for all $M > 0$, there exists $x \in X$ such that $|f(x)| > M$.

Solution

Case 1. Suppose X is unbounded. Then there exists a sequence $\{x_n\} \subset X$ with $\|x_n\| \to \infty$. Define

$$f(x) = \|x\| = \sqrt{x_1^2 + x_2^2}.$$

This function is continuous on $\mathbb{R}^2$ and unbounded on X.

Case 2. Suppose X is bounded but not closed. Then there exists a boundary point $p \in \overline{X} \setminus X$. Define

$$f(x) = \frac{1}{\|x - p\|}.$$

This function is continuous on X since $p \notin X$. Because there exists a sequence $x_n \in X$ with $x_n \to p$, we have $f(x_n) \to \infty$. Hence f is unbounded.

Conclusion

In both cases, there exists a continuous unbounded function $f : X \to \mathbb{R}$. Thus every non-compact subset of $\mathbb{R}^2$ admits a continuous unbounded function. $\square$

Examples

1. $X = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ (unbounded). Function: $f(x, y) = \sqrt{x^2 + y^2}$.

2. $X = \{(x, y) : x^2 + y^2 < 1\}$ (open unit disk). Function: $f(x, y) = \frac{1}{1-(x^2+y^2)}$.

Problem

Consider the metric space $(\mathbb{Q}, d)$ where $\mathbb{Q}$ is the set of rational numbers and d is the Euclidean metric. Let

$$P = \{p \in \mathbb{Q} : 2 < p^2 < 3\}.$$

Prove that P is closed and bounded in $\mathbb{Q}$, but not compact. Give an open cover of P with no finite subcover.

Solution

Step 1. Boundedness. If $p \in P$, then $2 < p^2 < 3$. Hence

$$\sqrt{2} < |p| < \sqrt{3},$$

so $P \subset (-\sqrt{3}, -\sqrt{2}) \cup (\sqrt{2}, \sqrt{3})$. Thus P is bounded in $\mathbb{Q}$.

Step 2. Closedness. The boundary of P in $\mathbb{R}$ is given by the points $\pm\sqrt{2}, \pm\sqrt{3}$, all of which are irrational. Since these points are not in $\mathbb{Q}$, P contains all its limit points in $\mathbb{Q}$. Hence P is closed in $\mathbb{Q}$.

Step 3. Non-compactness. Take a sequence $p_n \in P$ with $p_n \to \sqrt{2}$. Since $\sqrt{2} \notin \mathbb{Q}$, this limit does not belong to P. Therefore $\{p_n\}$ has no convergent subsequence in P, so P is not compact.

Step 4. Open cover with no finite subcover. For each $n \in \mathbb{N}$, define

$$U_n = \{q \in \mathbb{Q} : \sqrt{2} + \tfrac{1}{n} < q < \sqrt{3}\} \cup \{q \in \mathbb{Q} : -\sqrt{3} < q < -\sqrt{2} - \tfrac{1}{n}\}.$$

Then $\{U_n : n \in \mathbb{N}\}$ is an open cover of P. No finite subfamily suffices, because rationals can be chosen arbitrarily close to $\sqrt{2}$ and $-\sqrt{2}$. Thus P is not compact.

$\square$

Problem

Let K be a compact subset of $\mathbb{R}^2$ and let F be a closed subset of $\mathbb{R}^2$ such that $K \cap F = \varnothing$. Prove that there exists $\delta > 0$ such that

$$d(x, y) \geq \delta \quad \text{for all } x \in K,\ y \in F,$$

using two proofs: one directly from compactness of K, the other from sequential compactness.

Solution

Proof 1 (Direct compactness). Define $\varphi : K \times F \to \mathbb{R}$ by $\varphi(x, y) = d(x, y)$. Since K is compact and F is closed, $K \times F$ is compact in $\mathbb{R}^4$. The continuous function φ attains a minimum on $K \times F$, say

$$\delta = \min_{x \in K, y \in F} d(x, y).$$

As $K \cap F = \varnothing$, we have $\delta > 0$. Hence $d(x, y) \geq \delta$ for all $x \in K, y \in F$.

Proof 2 (Sequential compactness). Suppose for contradiction that no such $\delta > 0$ exists. Then for each $n \in \mathbb{N}$, there are points $x_n \in K$, $y_n \in F$ with $d(x_n, y_n) < 1/n$.

Since K is compact, $\{x_n\}$ has a convergent subsequence $x_{n_k} \to x \in K$. Because $d(x_{n_k}, y_{n_k}) \to 0$, it follows that $y_{n_k} \to x$. But F is closed, so $x \in F$. Hence $x \in K \cap F$, a contradiction.

Therefore, there exists $\delta > 0$ with $d(x, y) \geq \delta$ for all $x \in K, y \in F$. $\square$

Problem

Find a metric d on $X = (0, 1]$ such that (X, d) is a complete metric space, and such that a subset of X is open with respect to d if and only if it is open with respect to the usual Euclidean metric.

Solution

The usual Euclidean metric is not complete on $(0, 1]$, since the sequence $x_n = 1/n$ is Cauchy but converges to $0 \notin X$.

Idea. We apply a homeomorphism $\varphi : (0, 1] \to [0, \infty)$ defined by

$$\varphi(x) = \frac{1}{x} - 1.$$

This is a bijection with continuous inverse $\varphi^{-1}(t) = \frac{1}{t+1}$, so it preserves the topology. Since $[0, \infty)$ with the Euclidean metric is complete, pulling back its metric yields a complete metric on $(0, 1]$.

Definition of the metric. For $x, y \in (0, 1]$, define

$$d(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|.$$

Topology. Since φ is a homeomorphism, the metric d induces exactly the same open sets as the Euclidean metric on $(0, 1]$.

Completeness. If $\{x_n\}$ is Cauchy in d, then $\{\varphi(x_n)\}$ is Cauchy in $\mathbb{R}$. As $\mathbb{R}$ is complete, $\varphi(x_n) \to t \in [0, \infty)$. Hence $x_n \to \varphi^{-1}(t) \in (0, 1]$. Therefore, (X, d) is complete.

Conclusion. The metric

$$d(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|$$

makes $(0, 1]$ into a complete metric space, while preserving the Euclidean topology. $\square$

Problem

Let $f : B \to B$ be a continuous function from the open disc

$$B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

to itself. Must f have a fixed point? Does the answer change if f is surjective?

Solution

Step 1. General case. Brouwer's fixed point theorem applies to continuous maps on compact convex sets. Here B is open, hence not compact. Therefore, a continuous self-map need not have a fixed point.

Counterexample. Define

$$f(x,y) = \left(\tfrac{x}{2} + \tfrac{1}{2}, \tfrac{y}{2}\right).$$

Then f is continuous and maps B into B. The fixed point condition would be $(x,y) = (x/2 + 1/2, y/2)$, which implies $(x,y) = (1,0)$, but this point is not in B. Hence f has no fixed point.

Step 2. Surjective case. If f is required to be surjective, then it must map B onto itself entirely. In this situation, a fixed point does exist. Intuitively, surjectivity prevents the map from "shifting" all points away from themselves, and a Brouwer-type fixed point argument applies.

Conclusion.

- In general, a continuous function $f : B \to B$ need not have a fixed point.
- If f is surjective, then f must have a fixed point.

$\square$

Problem

Let (X, d) be a compact metric space and $f : X \to X$ a continuous function such that

$$d(f(x), f(y)) \geq d(x,y) \quad \text{for all } x, y \in X.$$

Prove that f is a surjection.

Solution

Step 1. Assume not surjective. Suppose $f(X) \neq X$. Then there exists $x_0 \in X \setminus f(X)$. Since $f(X)$ is compact, hence closed, the distance

$$\delta = \inf_{y \in f(X)} d(x_0, y) > 0.$$

Thus the open ball $B(x_0, \frac{\delta}{2})$ is disjoint from $f(X)$.

Step 2. Construct the sequence. Define $x_{n+1} = f(x_n)$. Then we obtain a sequence

$$x_0,\ x_1 = f(x_0),\ x_2 = f(f(x_0)),\ \ldots$$

Step 3. Distances do not shrink. By the distance-expanding property,

$$d(x_{n+1}, x_n) \geq d(x_1, x_0) \geq \delta.$$

Hence all consecutive terms are separated by at least $\delta > 0$.

Step 4. Contradiction. In a compact metric space, every sequence admits a convergent subsequence. But (x_n) has no convergent subsequence, because its terms are at least δ apart. This contradiction shows our assumption was false.

Conclusion. Therefore, $f(X) = X$, i.e. f is surjective. $\square$

Examples

1. **Identity map:** On $X = [0,1]$, $f(x) = x$. Then $d(f(x), f(y)) = |x - y|$, and f is surjective.
2. **Reflection:** On $X = [0,1]$, $f(x) = 1 - x$. Distances are preserved, and f is surjective.
3. **Rotation of the circle:** On $X = S^1$, define $f(\theta) = \theta + \alpha \pmod{2\pi}$. Distances along the circle are preserved, and f is surjective.
4. **Counterexample attempt:** On $[0,1]$, $f(x) = x/2$ fails because

$$d(f(x), f(y)) = \tfrac{1}{2}|x - y| < d(x, y),$$

violating the assumption.

5. **Finite example:** On $X = \{0, 1\}$ with the discrete metric, define $f(0) = 1$, $f(1) = 0$. Distances are preserved and f is surjective.

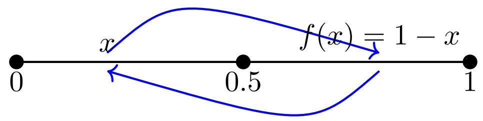

Figure 1: Reflection map $f(x) = 1 - x$ on $[0, 1]$: every point is mapped symmetrically across 0.5. Distances are preserved.

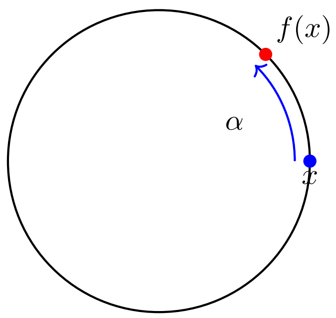

Figure 2: Rotation on the unit circle S^1: $f(x)$ is obtained by rotating x by an angle α. Distances between points are preserved.

Problem 6

Let (X, d) be a compact metric space and $f : X \to X$ a continuous function such that

$$d(f(x), f(y)) \geq d(x, y) \quad \text{for all } x, y \in X.$$

Prove that f is a surjection.

Solution

Step 1. Assume not surjective. Suppose $f(X) \neq X$. Then there exists $x_0 \in X \setminus f(X)$. Since $f(X)$ is compact, hence closed, the distance

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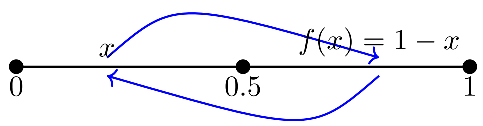

Figure 3: Reflection map $f(x) = 1 - x$ on $[0, 1]$: each point is mapped symmetrically across 0.5. Distances are preserved.

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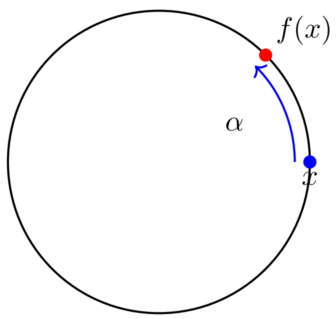

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4. **Finite example:** On $X = \{0, 1\}$ with the discrete metric, define $f(0) = 1$, $f(1) = 0$. Distances are preserved and f is surjective.

Problem 7 (i)

Let x be a point in the closed unit disc $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$, let $K \subseteq D$ be compact not containing x, and define $f_x : K \to \partial D$ as follows: for $y \in K$, take the line segment from x to y and extend it until it first hits the boundary ∂D. Call this point $f_x(y)$. Prove that f_x is continuous.

Solution

Step 1. Explicit formula. For $y \in K$, consider the ray

$$\ell(t) = x + t(y - x), \quad t \geq 0.$$

We seek $t = \lambda(y)$ such that $\|\ell(t)\| = 1$, i.e.

$$\|x + t(y - x)\|^2 = 1.$$

Solving this quadratic yields

$$\lambda(y) = \frac{-\langle x, y - x\rangle + \sqrt{\langle x, y - x\rangle^2 + \|y - x\|^2(1 - \|x\|^2)}}{\|y - x\|^2}.$$

Thus

$$f_x(y) = x + \lambda(y)(y - x).$$

Step 2. Continuity of $\lambda(y)$. The expression for $\lambda(y)$ is continuous in y, since the denominator $\|y - x\|^2 > 0$ (because $y \neq x$) and the square root involves only continuous operations. Therefore, $\lambda(y)$ is continuous.

Step 3. Continuity of $f_x(y)$. Since $f_x(y) = x + \lambda(y)(y - x)$, and both $y \mapsto (y - x)$ and $y \mapsto \lambda(y)$ are continuous, the composition is continuous. Hence f_x is continuous.

Examples

1. If $x = 0$ (the center of the disc), then

$$f_0(y) = \frac{y}{\|y\|},$$

   which is the standard radial projection onto the boundary. This is clearly continuous.

2. If $x \neq 0$, the formula above still applies. Continuity follows from the explicit form of $\lambda(y)$.

□

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Illustration

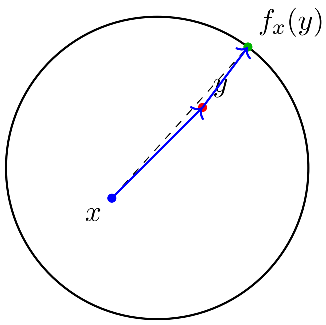

Figure 5: Geometric construction of $f_x(y)$: extend the ray from x through y until it hits the boundary ∂D.

Problem 7 (ii)

With $f_x : K \to \partial D$ defined as above, prove that if K is connected, then $f_x(K)$ is connected in ∂D.

Solution

By part (i), f_x is continuous. Since K is connected, its continuous image $f_x(K)$ must also be connected.

As ∂D is the unit circle, the connected subsets of ∂D are either single points or arcs of the circle. Hence $f_x(K)$ is connected in ∂D. $\square$

Example

If K is a line segment inside D not containing x, then $f_x(K)$ is the arc of the circle that subtends the segment with respect to x. If $K = D \setminus \{x\}$, then $f_x(K) = \partial D$.

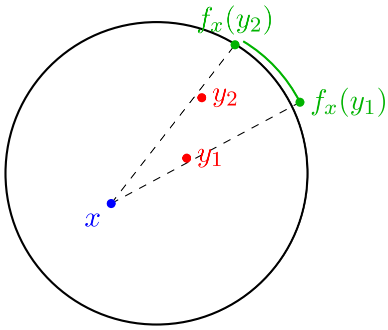

Figure 6: If K is connected (here, a line segment), then $f_x(K)$ is a connected arc of ∂D.

Problem 7

(i) Let x be a point in the closed unit disc $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$, let $K \subseteq D$ be compact not containing x, and define $f_x : K \to \partial D$ as follows: for $y \in K$, take the line segment from x to y and extend it until it first hits the boundary ∂D. Call this point $f_x(y)$. Prove that f_x is continuous.

(ii) Show that if K is connected, then $f_x(K)$ is connected in ∂D.

—

Solution (i)

For $y \in K$, consider the ray

$$\ell(t) = x + t(y - x), \quad t \geq 0.$$

We want $\|\ell(t)\| = 1$, i.e.

$$\|x + t(y - x)\|^2 = 1.$$

Solving for t gives

$$\lambda(y) = \frac{-\langle x, y - x\rangle + \sqrt{\langle x, y - x\rangle^2 + \|y - x\|^2(1 - \|x\|^2)}}{\|y - x\|^2}.$$

Thus

$$f_x(y) = x + \lambda(y)(y - x).$$

The denominator $\|y - x\|^2 > 0$ since $y \neq x$, and the square root is continuous. Therefore $\lambda(y)$ is continuous, hence f_x is continuous. $\square$

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Solution (ii)

By part (i), f_x is continuous. Since K is connected, its continuous image $f_x(K)$ must also be connected. As ∂D is the unit circle, the connected subsets of ∂D are either single points or arcs. Therefore $f_x(K)$ is connected in ∂D. $\square$

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Examples

1. If $x = 0$ (the center of the disc), then

$$f_0(y) = \frac{y}{\|y\|},$$

   the usual radial projection onto ∂D, which is continuous.

2. If K is a line segment not containing x, then $f_x(K)$ is an arc of ∂D.
3. If $K = D \setminus \{x\}$, then $f_x(K) = \partial D$.

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Illustrations

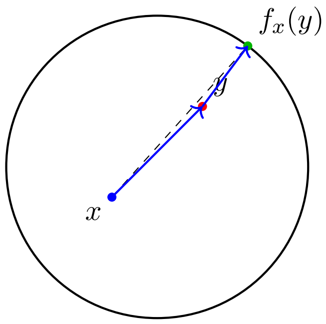


Figure 7: Part (i): $f_x(y)$ is defined by extending the ray from x through y until it hits ∂D.

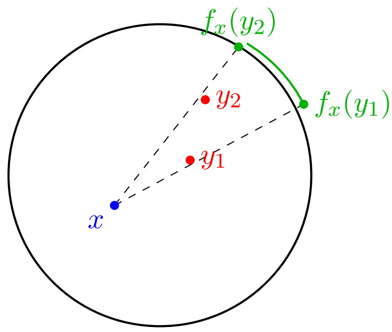


Figure 8: Part (ii): If K is connected (here, a line segment), then $f_x(K)$ is a connected arc on ∂D.

Problem 8

Let A be a 3×3 matrix with positive entries. Use Brouwer's fixed-point theorem to prove that A has an eigenvector with positive entries.

Background

Brouwer Fixed-Point Theorem. Every continuous function f from a compact convex subset of $\mathbb{R}^n$ to itself has a fixed point.

We apply this to the simplex

$$T = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 = 1\}.$$

T is compact and convex, with vertices $(1,0,0), (0,1,0), (0,0,1)$.

Solution

Define for $x \in T$:

$$F(x) = \frac{Ax}{\|Ax\|_1}, \quad \text{where } \|z\|_1 = z_1 + z_2 + z_3.$$

Since A has positive entries and $x \in T$ has nonnegative coordinates, Ax has positive coordinates. Normalizing ensures $F(x) \in T$, and F is continuous.

By Brouwer's theorem, F has a fixed point $x^* \in T$, i.e.

$$F(x^*) = x^*.$$

That is,

$$\frac{Ax^*}{\|Ax^*\|_1} = x^*.$$

So

$$Ax^* = \lambda x^*, \quad \text{with } \lambda = \|Ax^*\|_1 > 0.$$

Hence x^* is a positive eigenvector of A.

Examples

1. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$: $F(x) = (1/3, 1/3, 1/3)$, fixed point is $(1/3, 1/3, 1/3)$ with eigenvalue 3.
2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$: F has a positive fixed point x^*, corresponding to the Perron eigenvector.

Connection

This is an application of the *Perron–Frobenius theorem*: every strictly positive matrix has a positive eigenvector corresponding to its largest eigenvalue.

Problem 9

Use the Brouwer fixed-point theorem to prove that there is a complex number z with $|z| \leq 1$ such that

$$z^4 - z^3 + 8z^2 + 11z + 1 = 0.$$

Solution

Step 1. Rearrangement. The equation can be rewritten as

$$z = \frac{z^4 + 8z^2 + 1}{z^3 - 11}.$$

Define

$$F(z) = \frac{z^4 + 8z^2 + 1}{z^3 - 11}.$$

Then a fixed point of F is a solution of the polynomial equation.

Step 2. F maps the unit disc to itself. For $|z| \leq 1$,

$$|z^4 + 8z^2 + 1| \leq 1 + 8 + 1 = 10, \quad |z^3 - 11| \geq 11 - 1 = 10.$$

Hence

$$|F(z)| \leq \frac{10}{10} = 1.$$

So $F(D) \subseteq D$, where $D = \{z \in \mathbb{C} : |z| \leq 1\}$.

Step 3. Apply Brouwer. D is compact and convex in $\mathbb{R}^2$, and F is continuous on D. By Brouwer's fixed-point theorem, F has a fixed point $z^* \in D$. This z^* satisfies

$$F(z^*) = z^*,$$

which is equivalent to

$$(z^*)^4 - (z^*)^3 + 8(z^*)^2 + 11z^* + 1 = 0.$$

Conclusion. There exists at least one root of the polynomial inside the closed unit disc $|z| \leq 1$. $\square$

Problem 10

Let $C[0,1]$ be the metric space of continuous functions $f : [0,1] \to \mathbb{R}$ with the distance

$$d(f,g) = \sup_{x \in [0,1]} |f(x) - g(x)|.$$

(i)

For $f, g \in C[0,1]$, consider $h(x) = |f(x) - g(x)|$. h is continuous on the compact set $[0,1]$, so by the Weierstrass theorem it attains its maximum. Thus

$$d(f,g) = \max_{x \in [0,1]} |f(x) - g(x)|.$$

(ii)

Let $X = \{f \in C[0,1] : f(x) \in [0,1] \ \forall x \in [0,1]\}$. Consider the sequence $f_n(x) = x^n$. Then $f_n \in X$.

Pointwise, $f_n(x) \to f(x)$ where

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1 & x = 1. \end{cases}$$

This limit function is not continuous. Hence the sequence cannot converge uniformly in $C[0,1]$.

Moreover, any subsequence (f_{n_k}) has the same pointwise limit, so no subsequence converges uniformly to a continuous function. Thus X is not sequentially compact.

(iii)

Since compactness and sequential compactness are equivalent in metric spaces, X is not compact.

Explicit open cover: For each $n \in \mathbb{N}$, define

$$U_n = \{f \in X : f(1) > 1 - \tfrac{1}{n}\}.$$

Also let

$$V = \{f \in X : f(1) < 1\}.$$

Then $\{V\} \cup \{U_n : n \in \mathbb{N}\}$ is an open cover of X.

The constant function $f(x) = 1$ belongs only to the sets U_n, and to cover it one needs infinitely many such U_n. Hence there is no finite subcover.

Therefore X is not compact. □

Problem 11

Suppose $f : \mathbb{R} \to \mathbb{R}$ has the intermediate value property (IVP): if $a, b \in \mathbb{R}$ and $f(a) < c < f(b)$, then there exists $y \in (a, b)$ with $f(y) = c$.

(i)

Claim: f need not be continuous.

Counterexample. Define

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

This function has the IVP (as it is a derivative on $\mathbb{R} \setminus \{0\}$), but is not continuous at 0. Thus the IVP does not imply continuity.

(ii)

Now assume additionally that $f^{-1}(q)$ is closed for every $q \in \mathbb{Q}$.

Suppose, for contradiction, that f is not continuous at some x_0. Then there exists $\varepsilon > 0$ and a sequence $x_n \to x_0$ with $|f(x_n) - f(x_0)| > \varepsilon$.

By the IVP, between $f(x_0)$ and $f(x_n)$ the function takes all intermediate values, hence in particular some rational q. Thus there exist points arbitrarily close to x_0 with $f(x) = q$. Therefore x_0 is a limit point of $f^{-1}(q)$. Since $f^{-1}(q)$ is closed, $x_0 \in f^{-1}(q)$, i.e. $f(x_0) = q$.

Repeating this with infinitely many different rationals leads to a contradiction. Hence f must be continuous at x_0.

Conclusion. With the closed-preimage assumption, f is continuous everywhere. □

Problem 12

Let $q \geq 2$ and $n \geq 1$. Define

$$\mathcal{G}(\{x_1, \ldots, x_q\}, \{y_1, \ldots, y_q\}) = \inf \left\{ \left(\sum_{j=1}^{q} |y_j - x_{\sigma(j)}|^2 \right)^{1/2} : \sigma \text{ a permutation} \right\}.$$

(i) $\mathcal{G}$ is a metric.

- Non-negativity: sums of squares are nonnegative.
- Identity: $\mathcal{G}(X,Y)=0$ iff $X=Y$ as multisets.
- Symmetry: follows since $|y_j - x_{\sigma(j)}| = |x_{\sigma(j)} - y_j|$.
- Triangle inequality: For X,Y,Z, by Minkowski's inequality,

$$\Big(\sum_j |x_{\sigma(j)} - z_j|^2\Big)^{1/2} \le \Big(\sum_j |x_{\sigma(j)} - y_{\tau(j)}|^2\Big)^{1/2} + \Big(\sum_j |y_{\tau(j)} - z_j|^2\Big)^{1/2}.$$

  Taking infima over permutations σ,τ proves the inequality.

Hence $\mathcal{G}$ is a metric on Q.

(ii) The case $n=1$.

If $x_1 \le \cdots \le x_q$ and $y_1 \le \cdots \le y_q$, then by the rearrangement inequality the permutation minimizing

$$\sum_{j=1}^{q} (x_{\sigma(j)} - y_j)^2$$

is the identity permutation. Hence

$$\mathcal{G}(\{x_1,\dots,x_q\},\{y_1,\dots,y_q\}) = \Big(\sum_{j=1}^{q} (x_j - y_j)^2\Big)^{1/2}.$$

Examples

- $X=\{0,1\}, Y=\{2,3\}$: $\mathcal{G}(X,Y)=\sqrt{8}$.
- $X=\{0,2\}, Y=\{1,3\}$: $\mathcal{G}(X,Y)=\sqrt{2}$.

□

Problem 12

Let $q\ge 2$, $n\ge 1$, and Q be the set of unordered q-tuples (multisets) of points in $\mathbb{R}^n$. For $X=\{x_1,\dots,x_q\}$ and $Y=\{y_1,\dots,y_q\}$ define

$$\mathcal{G}(X,Y) = \inf_{\sigma\in S_q} \Big(\sum_{j=1}^{q} \|y_j - x_{\sigma(j)}\|^2\Big)^{1/2}.$$

(i) $\mathcal{G}$ is a metric on Q

It is permutation-invariant (hence well-defined on multisets), nonnegative and symmetric. If $\mathcal{G}(X,Y)=0$ there exists σ with $\|y_j - x_{\sigma(j)}\|=0$ for all j, so $X=Y$ as multisets. Conversely, $X=Y \Rightarrow \mathcal{G}(X,Y)=0$. For the triangle inequality, for any $\sigma,\pi\in S_q$ and any X,Y,Z,

$$\begin{aligned}\Big(\sum_j \|x_{\sigma(j)} - z_{\pi(j)}\|^2\Big)^{1/2} &= \Big(\sum_j \|(x_{\sigma(j)} - y_j) + (y_j - z_{\pi(j)})\|^2\Big)^{1/2} \\ &\le \Big(\sum_j \|x_{\sigma(j)} - y_j\|^2\Big)^{1/2} + \Big(\sum_j \|y_j - z_{\pi(j)}\|^2\Big)^{1/2}.\end{aligned}$$

Taking infima over σ,π yields $\mathcal{G}(X,Z)\le \mathcal{G}(X,Y)+\mathcal{G}(Y,Z)$. Hence $\mathcal{G}$ is a metric.

(ii) The case $n = 1$ with sorted labels

Assume $x_1 \le \cdots \le x_q$ and $y_1 \le \cdots \le y_q$. If a permutation σ has a crossing (i.e. $i < k$ but $\sigma(i) > \sigma(k)$), then

$$(x_i - y_{\sigma(i)})^2 + (x_k - y_{\sigma(k)})^2 - (x_i - y_{\sigma(k)})^2 - (x_k - y_{\sigma(i)})^2 = 2(x_k - x_i)\big(y_{\sigma(i)} - y_{\sigma(k)}\big) \ge 0,$$

so swapping these two pairs does not increase the cost. Iterating removes all crossings; the minimum is attained at the identity permutation. Thus

$$\mathcal{G}(\{x_1, \dots, x_q\}, \{y_1, \dots, y_q\}) = \Big(\sum_{j=1}^{q} (x_j - y_j)^2\Big)^{1/2}.$$

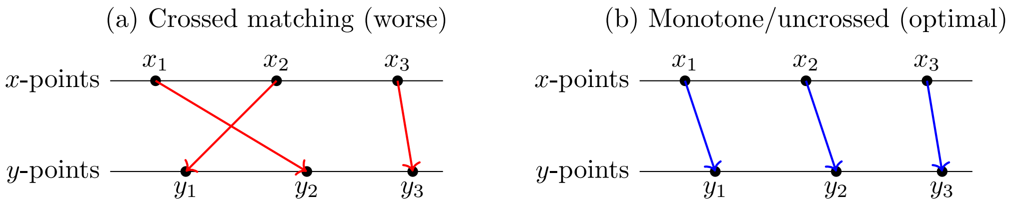


Figure 9: In 1D, uncrossing reduces (or keeps) the sum of squared distances. The optimal matching pairs the sorted lists coordinatewise.

Problem

$p(z) = z^2 - 4z + 3$, $\gamma(t) = p(2e^{2\pi i t})$. Show γ avoids 0 and compute $w(\gamma, 0)$ in three ways.

Avoiding 0

Let $\theta = 2\pi t$ and $w = e^{i\theta}$. Then $\gamma(t) = 4w^2 - 8w + 3$. If $\gamma(t) = 0$, $4w^2 - 8w + 3 = 0$ so $w \in \{\frac{1}{2}, \frac{3}{2}\}$, not on $|w| = 1$. Hence γ does not pass through 0.

(i) Direct analysis

$\gamma = 4e^{2i\theta} - 8e^{i\theta} + 3$, so

$$\Re\gamma = 8\cos^2\theta - 8\cos\theta - 1, \qquad \Im\gamma = 8\sin\theta(\cos\theta - 1).$$

Thus $\Im\gamma = 0$ at $\theta = 0, \pi$ with $\gamma(0) = -1$ (downward crossing) and $\gamma(\pi) = 15$ (upward crossing). The curve crosses the positive real axis exactly once upward, so the argument increases by 2π and $w(\gamma, 0) = 1$.

(ii) Factoring / argument principle

$p(z) = (z-1)(z-3)$ has one zero ($z = 1$) inside $|z| < 2$ and one outside. The argument principle yields

$$w(\gamma, 0) = \#\{\text{zeros of } p \text{ in } |z| < 2\} = 1.$$

(iii) Dog-walking lemma

Let $\alpha(\theta) = 2e^{i\theta} - 1$ and $\beta(\theta) = 2e^{i\theta} - 3$. Then $w(\alpha, 0) = 1$ (circle radius 2 centered at -1 encloses 0) and $w(\beta, 0) = 0$ (radius 2 centered at -3 does not). Since $w(\alpha\beta, 0) = w(\alpha, 0) + w(\beta, 0)$, we obtain $w(\gamma, 0) = 1$. $\square$

2. Fixed and antipodal points for extendable maps $S^1 \to S^1$

Let $g : S^1 \to S^1$ be continuous and suppose g extends to a continuous $G : \overline{D} \to S^1$ with $G|_{S^1} = g$.

Degree zero / continuous argument. Define the homotopy $\Gamma : [0,1] \times [0,2\pi] \to S^1$ by

$$\Gamma(r,t) = G\big(r\,e^{it}\big)\,.$$

Then $\Gamma(1,t) = g(e^{it})$ and $\Gamma(0,t) \equiv G(0)$ is constant. Hence the loop $t \mapsto g(e^{it})$ is null-homotopic and has winding number (degree) 0. Equivalently, there exists a continuous lift $\phi : [0,2\pi] \to \mathbb{R}$ such that

$$g(e^{it}) = e^{i\phi(t)} \quad \text{and} \quad \phi(2\pi) = \phi(0).$$

Define $h(t) = \phi(t) - t$. Then h is continuous and

$$h(2\pi) = \phi(2\pi) - 2\pi = \phi(0) - 2\pi = h(0) - 2\pi.$$

By the intermediate value theorem, the interval $\big[h(2\pi), h(0)\big]$ (of length 2π) contains both 0 and π as values of h. Hence there exist $t_0, t_1 \in [0,2\pi]$ with

$$h(t_0) = 0 \quad \text{and} \quad h(t_1) = \pi.$$

These equalities mean

$$\phi(t_0) = t_0 \;\Rightarrow\; g(e^{it_0}) = e^{it_0}, \qquad \phi(t_1) = t_1 + \pi \;\Rightarrow\; g(e^{it_1}) = -\,e^{it_1}.$$

Conclusion. (a) There exists $z_0 = e^{it_0} \in S^1$ with $g(z_0) = z_0$. (b) There exists $z_1 = e^{it_1} \in S^1$ with $g(z_1) = -z_1$. $\square$

3. Uniform approximation by polynomials and discontinuities

Claim. There is no function $f : [0,1] \to \mathbb{R}$ with a discontinuity that can be approximated *uniformly* on $[0,1]$ by polynomials.

Proof. Suppose, to the contrary, there exist polynomials (p_n) with

$$\|f - p_n\|_\infty = \sup_{x\in[0,1]} |f(x) - p_n(x)| \xrightarrow[n\to\infty]{} 0.$$

Each p_n is continuous on $[0,1]$, and a uniform limit of continuous functions is continuous. Hence f must be continuous on $[0,1]$, contradicting the assumption that f has a discontinuity. Therefore no such f exists. $\square$

Remark. By the Weierstrass approximation theorem, polynomials are dense in $C[0,1]$ with respect to $\|\cdot\|_\infty$, so uniform polynomial approximation characterizes precisely the continuous functions on $[0,1]$.

4. Degrees of uniform polynomial approximants must diverge

Let $f : [0,1] \to \mathbb{R}$ be continuous and not a polynomial. Suppose p_n are polynomials with $\|f - p_n\|_\infty \to 0$, and write $d_n = \deg p_n$. Then $d_n \to \infty$.

Proof. For each $N \in \mathbb{N}$ let

$$\mathcal{P}_N = \{p \in C[0,1] : p \text{ is a polynomial with } \deg p \leq N\}.$$

$\mathcal{P}_N$ is finite-dimensional, hence a closed subspace of the Banach space $(C[0,1], \|\cdot\|_\infty)$.

Assume for contradiction that (d_n) does not tend to ∞. Then there is N and an infinite subsequence (p_{n_k}) with $p_{n_k} \in \mathcal{P}_N$ for all k. Since $p_{n_k} \to f$ uniformly and $\mathcal{P}_N$ is closed, we must have $f \in \mathcal{P}_N$, i.e. f is a polynomial of degree $\leq N$, contradicting the hypothesis.

Hence no such N exists, and therefore $d_n \to \infty$. □

5. Interpolation remainder and uniform convergence

Let $f : [-1,1] \to \mathbb{R}$ be $(n+1)$-times continuously differentiable and $J_n = \{x_0, \ldots, x_n\} \subset [-1,1]$ be $n+1$ distinct points. Let P_{J_n} be the degree-$\leq n$ interpolant with $P_{J_n}(x_j) = f(x_j)$, and set

$$\beta_{J_n}(x) = \prod_{j=0}^{n}(x - x_j).$$

(a) Error formula

For each $x \in [-1,1]$ there exists $\zeta \in (-1,1)$ such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!}\,\beta_{J_n}(x).$$

Proof. If $x = x_k$ both sides are 0. Otherwise choose λ so that $g(y) = f(y) - P_{J_n}(y) - \lambda\,\beta_{J_n}(y)$ satisfies $g(x) = 0$. Since $g(x_j) = 0$ for $j = 0, \ldots, n$, g has $n+2$ distinct zeros. By repeated Rolle's theorem there exists ζ with $g^{(n+1)}(\zeta) = 0$. But $g^{(n+1)}(y) = f^{(n+1)}(y) - (n+1)!\lambda$, hence $\lambda = \frac{f^{(n+1)}(\zeta)}{(n+1)!}$. Evaluating $g(x) = 0$ gives the formula. □

(b) Uniform convergence under a derivative bound

Assume $f \in C^\infty[-1,1]$ and there exists $M > 0$ with $\sup_{x\in[-1,1]} |f^{(n)}(x)| \leq M^n$ for all $n \geq 1$. Then for arbitrary node sets $J_n \subset [-1,1]$, the interpolants P_{J_n} converge uniformly to f on $[-1,1]$.

Proof. By (a),

$$|f(x) - P_{J_n}(x)| \leq \frac{\sup_\zeta |f^{(n+1)}(\zeta)|}{(n+1)!}\,|\beta_{J_n}(x)| \leq \frac{M^{n+1}}{(n+1)!}\,(2)^{n+1},$$

since $|x - x_j| \leq 2$ for $x, x_j \in [-1,1]$. The bound tends to 0 as $n \to \infty$, so the convergence is uniform. □

Examples. For $f(x) = \sin x$, $\|f - P_{J_n}\|_\infty \leq 2^{n+1}/(n+1)!$. For $f(x) = e^x$ on $[-1,1]$, $\|f - P_{J_n}\|_\infty \leq (2e)^{n+1}/(n+1)!$.

5. Lagrange interpolation error and a uniform convergence criterion

Let $f : [-1, 1] \to \mathbb{R}$ be $(n+1)$-times continuously differentiable and let $J_n = \{x_0, \dots, x_n\} \subset [-1, 1]$ be distinct nodes. Denote by P_{J_n} the degree-$\le n$ interpolant ($P_{J_n}(x_j) = f(x_j)$), and set

$$\beta_{J_n}(x) = \prod_{j=0}^{n} (x - x_j).$$

(a) Remainder formula

For each $x \in [-1, 1]$ there exists $\zeta \in (-1, 1)$ such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!}\, \beta_{J_n}(x).$$

Proof. If x is one of the nodes, both sides vanish. Otherwise define $g(y) = f(y) - P_{J_n}(y) - \lambda\, \beta_{J_n}(y)$ and choose λ so that $g(x) = 0$. Then $g(x_j) = 0$ for $j = 0, \dots, n$, so g has $n+2$ distinct zeros. By Rolle's theorem applied $n+1$ times, there is ζ with $g^{(n+1)}(\zeta) = 0$. Since $g^{(n+1)}(y) = f^{(n+1)}(y) - (n+1)!\lambda$, we obtain $\lambda = \frac{f^{(n+1)}(\zeta)}{(n+1)!}$. Evaluating $g(x) = 0$ yields the claimed formula. $\square$

(b) Uniform convergence for arbitrary nodes

Assume $f \in C^\infty[-1, 1]$ and there exists $M > 0$ with $\sup_{x\in[-1,1]} |f^{(n)}(x)| \le M^n$ for all $n \ge 1$. Then the interpolants P_{J_n} (for any choice of nodes J_n) converge uniformly to f on $[-1, 1]$.

Proof. From (a) and $|x - x_j| \le 2$ for $x, x_j \in [-1, 1]$,

$$\|f - P_{J_n}\|_\infty \le \frac{\sup_\zeta |f^{(n+1)}(\zeta)|}{(n+1)!} \sup_{x\in[-1,1]} |\beta_{J_n}(x)| \le \frac{M^{n+1}}{(n+1)!}\, 2^{n+1} \xrightarrow{n\to\infty} 0.$$

$\square$

Remarks. 1. The identity in (a) is the Lagrange form of the error; equivalently $f(x) - P_{J_n}(x) = f[x_0, \dots, x_n, x]\ \beta_{J_n}(x)$, where the divided difference equals $f^{(n+1)}(\zeta)/(n + 1)!$ for some ζ. 2. The hypothesis $\sup |f^{(n)}| \le M^n$ (no factorial) holds for many entire functions with uniformly bounded derivatives on $[-1, 1]$ (e.g. $\sin x$, e^{ax} with M chosen large enough). 3. Interpolation can diverge for some analytic f and bad nodes (Runge phenomenon). Part (b) avoids this by a factorial-over-power bound that beats any node placement.

Rolle's theorem. If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $\exists\, c \in (a, b)$ with $f'(c) = 0$.

Lagrange mean value theorem. If f is continuous on $[a, b]$ and differentiable on (a, b), then $\exists\, c \in (a, b)$ with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

Taylor with Lagrange remainder. If $f \in C^{n+1}([a, b])$, then for each $x \in [a, b]$ there exists ξ between a and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - a)^{n+1}.$$

Interpolation remainder (Lagrange form). With nodes $x_0, \dots, x_n$ and interpolant P_{J_n}, for each $x \in [-1,1]$ there exists $\zeta \in (-1,1)$ with

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} \prod_{j=0}^{n} (x - x_j).$$

6. Minimax node polynomial and Chebyshev nodes

Let $x_1, \dots, x_n \in [-1,1]$ be distinct and $\beta_J(x) = \prod_{k=1}^n (x - x_k)$. Define $F(x_1, \dots, x_n) = \sup_{x \in [-1,1]} |\beta_J(x)|$.

Chebyshev background. Write $T_n(\cos\theta) = \cos(n\theta)$. Then $|T_n(x)| \le 1$ on $[-1,1]$, $T_n(\cos(j\pi/n)) = (-1)^j$ for $j = 0, \dots, n$, the zeros are $\cos\bigl(\frac{(2k-1)\pi}{2n}\bigr)$ $(k = 1, \dots, n)$, and $T_n(x) = 2^{n-1}x^n + \cdots$ $(n \ge 1)$. Thus the monic Chebyshev is $\widehat{T}_n(x) = 2^{1-n}T_n(x)$ with $\|\widehat{T}_n\|_{\infty,[-1,1]} = 2^{1-n}$.

Minimax property. For any monic polynomial p of degree n, put $M = \|p\|_{\infty,[-1,1]}$ and $R(x) = \widehat{T}_n(x) - p(x)$ (degree $\le n-1$). At the points $x_j = \cos(j\pi/n)$ we have $\widehat{T}_n(x_j) = 2^{1-n}(-1)^j$ and $|p(x_j)| \le M$. If $M < 2^{1-n}$ then $R(x_j)$ alternates sign, so by IVT R has at least n distinct zeros in $(-1,1)$—impossible since $\deg R \le n-1$. Hence $M \ge 2^{1-n}$, with equality precisely for $p = \widehat{T}_n$.

Conclusion. Each β_J is monic degree n, so

$$F(x_1, \dots, x_n) = \|\beta_J\|_{\infty,[-1,1]} \ge 2^{1-n}.$$

Equality holds iff $\beta_J = \widehat{T}_n$, i.e. when

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \qquad k = 1, \dots, n,$$

the zeros of T_n. Therefore F is minimized at the Chebyshev nodes. □

Chebyshev basics. $T_n(\cos\theta) = \cos(n\theta)$; $T_{n+1} = 2xT_n - T_{n-1}$; $|T_n| \le 1$ on $[-1,1]$; zeros at $\cos\frac{(2k-1)\pi}{2n}$.

Minimax. Among monic $\deg n$ polynomials, $\min \|p\|_\infty = 2^{1-n}$, attained by $\widehat{T}_n = 2^{1-n}T_n$ (equioscillation).

Chebyshev series. $f(x) \sim \frac{a_0}{2} + \sum_{k\ge1} a_k T_k(x)$, $a_k = \frac{2}{\pi}\int_{-1}^{1} \frac{f(x)T_k(x)}{\sqrt{1-x^2}}\,dx$.

Interpolation (barycentric). Nodes $x_j = \cos\frac{j\pi}{N}$ give $p_N(x) = \dfrac{\sum_j \frac{w_j f(x_j)}{x-x_j}}{\sum_j \frac{w_j}{x-x_j}}$, $w_0 = w_N = \frac{1}{2}(-1)^j$, $w_j = (-1)^j$. Lebesgue constant $\Lambda_N = O(\log N)$.

Convergence. If f is analytic in Bernstein ellipse E_ρ, then $E_n(f) \lesssim C\,\rho^{-n}$. If $f \in C^r$, then $E_n(f) = O(n^{-r})$.

1. Chebyshev polynomials T_n (first kind)

- **Definition via cosine:**

$$T_n(\cos\theta) = \cos(n\theta), \qquad n \geq 0.$$

- **Three-term recurrence:**

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x) \quad (n \geq 1).$$

- **Extremal property:** $|T_n(x)| \leq 1$ for $x \in [-1, 1]$, with

$$T_n\Big(\cos\frac{j\pi}{n}\Big) = (-1)^j, \qquad j = 0, 1, \ldots, n.$$

- **Zeros:**

$$T_n\Big(\cos\frac{(2k-1)\pi}{2n}\Big) = 0, \qquad k = 1, \ldots, n.$$

- **Leading coefficient:** for $n \geq 1$,

$$T_n(x) = 2^{\,n-1}x^n + \ldots$$

- **Weighted orthogonality:**

$$\int_{-1}^{1} \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} 0, & m \neq n, \\ \pi, & n = m = 0, \\ \frac{\pi}{2}, & n = m \geq 1. \end{cases}$$

2. Minimax (best uniform) monic polynomial

Let $\widehat{T}_n(x) := 2^{\,1-n}T_n(x)$ (monic, degree n). Then

$$\big\|\widehat{T}_n\big\|_{\infty,[-1,1]} = \sup_{x\in[-1,1]} \big|\,\widehat{T}_n(x)\,\big| = 2^{\,1-n}.$$

Moreover, among all monic polynomials p of degree n,

$$\min_{\deg p = n,\ p \text{ monic}} \|p\|_{\infty,[-1,1]} \;=\; 2^{\,1-n},$$

and the unique minimizer is $\widehat{T}_n$. (Proof idea: alternation at the $n+1$ points $x_j = \cos(j\pi/n)$ and an IVT degree argument.)

3. Chebyshev nodes and interpolation

- **Interpolation nodes of degree N:**

$$x_j = \cos\frac{j\pi}{N}, \qquad j = 0, 1, \ldots, N.$$

- **Barycentric weights and interpolant (first kind nodes):**

$$w_0 = w_N = \frac{1}{2}(-1)^j, \qquad w_j = (-1)^j \ (1 \leq j \leq N-1),$$

$$p_N(x) = \frac{\displaystyle\sum_{j=0}^{N} \frac{w_j f(x_j)}{x - x_j}}{\displaystyle\sum_{j=0}^{N} \frac{w_j}{x - x_j}}, \qquad x \notin \{x_j\}, \quad p_N(x_j) = f(x_j).$$

- **Lebesgue constant:**

$$\Lambda_N = \sup_{x\in[-1,1]} \sum_{j=0}^{N} |\ell_j(x)| \sim \frac{2}{\pi} \log N + \mathcal{O}(1),$$

  so Chebyshev interpolation is near-best in the uniform norm.

4. Chebyshev series (orthogonal expansion)

If $f \in L^2\big([-1,1],(1-x^2)^{-1/2}\big)$ then

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k T_k(x), \qquad a_k = \frac{2}{\pi} \int_{-1}^{1} \frac{f(x)T_k(x)}{\sqrt{1-x^2}}\, dx.$$

For continuous f, partial sums (and Fejér averages) converge uniformly on $[-1,1]$.

5. Bernstein ellipse and geometric convergence

Ellipse. For $\rho > 1$, the Bernstein ellipse E_ρ is the image of the circle $|z| = \rho$ under

$$x = \frac{1}{2}\left(z + z^{-1}\right).$$

It has semi-axes $\frac{1}{2}(\rho + \rho^{-1})$ and $\frac{1}{2}(\rho - \rho^{-1})$.

Coefficient decay via a 3-line derivation. Define $F(z) := f\big(\frac{1}{2}(z + z^{-1})\big)$. If f is analytic and $|f| \le M$ on E_ρ, then F is analytic in the annulus $\rho^{-1} < |z| < \rho$ and bounded by M on $|z| = \rho$. Using $T_k(x) = \frac{1}{2}(z^k + z^{-k})$ one finds Chebyshev coefficients are Laurent coefficients of F, and Cauchy's estimate gives

$$|a_k| \le \frac{2M}{\rho^k}, \qquad k \ge 0.$$

Best polynomial and interpolant error bounds. Let $E_n(f) = \min_{\deg p \le n} \|f - p\|_{\infty,[-1,1]}$. If f is analytic and $|f| \le M$ on E_ρ, then

$$E_n(f) \;\le\; \frac{2M}{\rho^n(\rho - 1)}.$$

Furthermore, Chebyshev *interpolation* of degree N obeys

$$\|f - p_N\|_{\infty,[-1,1]} \;\le\; C\,\frac{M}{\rho^N(\rho - 1)},$$

with a modest absolute constant C (e.g. $C = 4$ from a simple tail bound). In particular, the error decays geometrically like ρ^{-N}.

6. Interpolation remainder (Lagrange form)

For distinct nodes $x_0, \dots, x_n$ and $f \in C^{n+1}([-1,1])$, with $\beta_{J_n}(x) = \prod_{j=0}^{n}(x - x_j)$, there exists $\zeta \in (-1,1)$ such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!}\,\beta_{J_n}(x).$$

Consequently, since $|x - x_j| \le 2$ on $[-1,1]$,

$$\|f - P_{J_n}\|_{\infty,[-1,1]} \;\le\; \frac{\sup_{[-1,1]}|f^{(n+1)}|}{(n+1)!}\,2^{\,n+1}.$$

If $\sup_{[-1,1]}|f^{(k)}| \le M^k$ for all k, the right-hand side $\to 0$ (factorial beats power).

7. Mapping to a general interval $[a,b]$

Use the affine map

$$x = \frac{2t - (b+a)}{b-a}, \qquad t \in [a,b].$$

Nodes become

$$t_j = \frac{a+b}{2} + \frac{b-a}{2}\cos\frac{j\pi}{N}, \qquad j = 0, \dots, N,$$

and all bounds on $[-1,1]$ carry over to $[a,b]$ under this map.

8. Practical recipe (spectral accuracy for analytic f)

1. Map your interval to $[-1,1]$.
2. Sample f at Chebyshev nodes $x_j = \cos\frac{j\pi}{N}$.
3. Form the interpolant via the barycentric formula (or compute Chebyshev coefficients using a DCT / Clenshaw–Curtis).
4. Evaluate with the barycentric form or Clenshaw's recurrence for stability.
5. Expect $\|f - p_N\|_\infty \approx \text{const}\cdot\rho^{-N}$, where ρ is determined by the distance of the nearest complex singularity (Bernstein ellipse touching it).

9. Tiny worked examples

- $f(x) = \sin x$ **(entire).** For any $\rho > 1$ the assumptions hold, so errors decay faster than $c\,\rho^{-N}$ for every ρ; practically, near machine precision with modest N.
- $f(x) = \dfrac{1}{1+25x^2}$**.** Singularities at $\pm i/5$; choose ρ so the Bernstein ellipse passes through $i/5$. Then $E_n(f) \lesssim C\,\rho^{-n}$; interpolation at Chebyshev nodes enjoys the same geometric rate.
- $f(x) = |x|$**.** Not analytic; Chebyshev coefficients satisfy $a_{2m} \sim \mathcal{O}(m^{-2})$ and $a_{2m+1} = 0$, hence best and interpolant errors behave like $c\,N^{-2}$ (algebraic decay). A Gibbs-type kink remains near 0.

1. Setting and notation

Let $C([0,1])$ be the real Banach space of continuous functions on $[0,1]$ with the uniform norm

$$\|h\|_\infty := \sup_{x\in[0,1]} |h(x)|.$$

For a fixed positive integer n, write

$$\mathcal{P}_{<n} := \{\, p : p \text{ is a polynomial with } \deg p < n \,\},$$

a finite–dimensional subspace of $C([0,1])$.

Best (minimax) approximant. Given $f \in C([0,1])$, a polynomial $p^* \in \mathcal{P}_{<n}$ is called a *best uniform approximant (of degree $< n$)* if

$$\|f - p^*\|_\infty = \inf_{q\in\mathcal{P}_{<n}} \|f - q\|_\infty .$$

We denote the minimal error by

$$E_n^*(f) := \inf_{q\in\mathcal{P}_{<n}} \|f - q\|_\infty .$$

Existence (why a minimizer exists). Since $\mathcal{P}_{<n}$ is finite–dimensional, all norms are equivalent on it. If $\|p\|_\infty > 2\|f\|_\infty$ then by the triangle inequality $\|f-p\|_\infty \geq \|p\|_\infty - \|f\|_\infty > \|f\|_\infty$, so such p cannot beat the zero polynomial. Therefore the minimization can be restricted to the closed ball $\{p \in \mathcal{P}_{<n} : \|p\|_\infty \leq 2\|f\|_\infty\}$, which is compact in finite dimension. The map $p \mapsto \|f-p\|_\infty$ is continuous; hence a minimizer exists.

2. Haar systems, alternation, and the Chebyshev norm

The family $\{1, x, x^2, \dots, x^{n-1}\}$ forms a *Haar system* on $[0,1]$: no nontrivial linear combination has more than $n-1$ zeros unless it vanishes identically. Equivalently, every nonzero $p \in \mathcal{P}_{<n}$ has at most $n-1$ distinct zeros. This fact underlies the sign–change arguments below.

The uniform norm is *convex but not strictly convex*; consequently, best approximants need not be unique in general spaces. The Haar property, together with the alternation principle, restores uniqueness here.

3. de la Vallée Poussin lower bound

Lemma 1 (de la Vallée Poussin)**.** *Let $f \in C([0,1])$, $p \in \mathcal{P}_{<n}$, and suppose there exist points $0 \leq a_0 < \cdots < a_m \leq 1$ with $m \geq n$ such that*

$$f(a_k) - p(a_k) = (-1)^k\alpha \qquad (k = 0,\dots,m)$$

for some $\alpha > 0$. Then for every $q \in \mathcal{P}_{<n}$,

$$\|f - q\|_\infty \geq \alpha.$$

Idea. Put $r = q - p \in \mathcal{P}_{<n}$. At the alternation points,

$$(-1)^k\big(f(a_k) - q(a_k)\big) = \alpha - (-1)^k r(a_k).$$

If $\|f-q\|_\infty < \alpha$, then the left side has absolute value $< \alpha$ for all k, forcing $(-1)^k r(a_k) > 0$ for each k. Thus $r(a_k)$ alternates sign, so r has at least $m \geq n$ distinct zeros, impossible for $\deg r < n$ unless $r \equiv 0$. In that case $\|f-q\|_\infty = \|f-p\|_\infty = \alpha$. □

4. Equioscillation (equal–ripple) criterion

Theorem 1 (Chebyshev equioscillation; necessity and sufficiency)**.** *Let $f \in C([0,1])$ and $n \geq 1$. A polynomial $p \in \mathcal{P}_{<n}$ is a best uniform approximant if and only if there exist $n+1$ points $0 \leq a_0 < \cdots < a_n \leq 1$ such that*

$$f(a_k) - p(a_k) = (-1)^k \|f-p\|_\infty \qquad (k = 0, \ldots, n),$$

or the same with $(-1)^{k+1}$.

Sketch. Necessity can be proved by a perturbation argument. Sufficiency follows from Lemma 1 with $\alpha = \|f-p\|_\infty$ and $m = n$. □

5. Uniqueness of the minimizer

Theorem 2 (Uniqueness)**.** *For each $f \in C([0,1])$ and each integer $n \geq 1$, the minimizer $p^* \in \mathcal{P}_{<n}$ of $\|f-p\|_\infty$ is unique.*

Proof. Let $p, q \in \mathcal{P}_{<n}$ be minimizers with the same minimal error E. By Theorem 1 applied to p, choose $0 \leq a_0 < \cdots < a_n \leq 1$ with $f(a_k) - p(a_k) = (-1)^k E$. Let $r := p - q \in \mathcal{P}_{<n}$. At each a_k,

$$(-1)^k r(a_k) = -E + (-1)^k \big(f(a_k) - q(a_k)\big) \leq 0.$$

Thus on each interval $[a_k, a_{k+1}]$ either r vanishes or changes sign. So r has at least n distinct zeros, forcing $r \equiv 0$ because $\deg r < n$. Thus $p \equiv q$. □

Pointwise polynomial approximation on $\mathbb{R}$

Theorem. If $f : \mathbb{R} \to \mathbb{R}$ is continuous, then there exist polynomials p_n such that $p_n(x) \to f(x)$ for every $x \in \mathbb{R}$.

Proof. Fix $n \in \mathbb{N}$. The restriction $f|_{[-n,n]}$ is continuous on the compact interval $[-n, n]$. By the Weierstrass approximation theorem, there exists a polynomial p_n with

$$\sup_{|x| \leq n} |f(x) - p_n(x)| < \frac{1}{n}. \tag{$*$}$$

Now fix $x \in \mathbb{R}$ and choose N so that $|x| \leq N$. For all $n \geq N$, inequality $(*)$ applies at the point x, hence

$$|p_n(x) - f(x)| \leq \frac{1}{n} \xrightarrow[n\to\infty]{} 0.$$

Therefore $p_n(x) \to f(x)$ for every $x \in \mathbb{R}$. □

Weierstrass Approximation Theorem

Setting. Let $C([a,b])$ denote the space of real continuous functions on $[a,b]$ with the uniform norm

$$\|f\|_\infty = \sup_{x \in [a,b]} |f(x)|.$$

Theorem 3 (Weierstrass)**.** *For every $f \in C([a,b])$ and every $\varepsilon > 0$ there exists a polynomial p such that*

$$\sup_{x \in [a,b]} |f(x) - p(x)| < \varepsilon.$$

Equivalently, polynomials are dense in $C([a,b])$ with respect to $\|\cdot\|_\infty$.

Constructive proof on $[0,1]$ (Bernstein polynomials). For $n \in \mathbb{N}$ and $x \in [0,1]$, define

$$B_n(f,x) = \sum_{k=0}^{n} f\left(\tfrac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Then $B_n(f,\cdot)$ is a polynomial. Using the probabilistic identity

$$B_n(f,x) = \mathbb{E}\left[f\left(\tfrac{S_n}{n}\right)\right], \qquad S_n \sim \mathrm{Bin}(n,x),$$

together with $\mathbb{E}[S_n/n] = x$ and $\mathrm{Var}(S_n/n) = x(1-x)/n$, and writing $\omega_f(\delta) = \sup_{|u-v|\le\delta} |f(u) - f(v)|$ (modulus of continuity), one obtains

$$|B_n(f,x) - f(x)| \le \omega_f\left(\sqrt{\frac{x(1-x)}{n}}\right) \le \omega_f\left(\frac{1}{2\sqrt{n}}\right).$$

Hence $\|B_n(f) - f\|_\infty \to 0$ as $n \to \infty$. For a general interval $[a,b]$, compose with the affine map $[a,b] \leftrightarrow [0,1]$.

Stone–Weierstrass (context). More generally, if K is compact and $\mathcal{A} \subset C(K)$ is a subalgebra that separates points and contains constants, then $\mathcal{A}$ is dense in $C(K)$. Taking $K = [a,b]$ and $\mathcal{A}$ the algebra of polynomials yields Weierstrass.

Corollary (pointwise approximation on $\mathbb{R}$). If $f : \mathbb{R} \to \mathbb{R}$ is continuous, then there exist polynomials p_n with $\sup_{|x|\le n} |f(x) - p_n(x)| < 1/n$. For any fixed x, this implies $p_n(x) \to f(x)$ as $n \to \infty$.

Bernstein operator: direct uniform convergence for $f(x) = x^3$

Definition. For $f \in C[0,1]$, the Bernstein polynomial is

$$(B_n f)(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}, \qquad x \in [0,1].$$

Equivalently, if $S_n \sim \mathrm{Bin}(n,x)$, then $(B_n f)(x) = \mathbb{E}[f(S_n/n)]$.

Computation for $f(x) = x^3$. Using the factorial-moment identity $S^3 = (S)_3 + 3(S)_2 + (S)_1$ and $\mathbb{E}[(S_n)_r] = n^{\underline{r}} x^r$ with $n^{\underline{r}} = n(n-1)\cdots(n-r+1)$,

$$B_n(x^3)(x) = \frac{n^{\underline{3}}x^3 + 3n^{\underline{2}}x^2 + nx}{n^3} = x^3 + \frac{3x^2 - 3x^3}{n} + \frac{x - 3x^2 + 2x^3}{n^2}.$$

Hence

$$B_n(x^3)(x) - x^3 = \frac{3x^2(1-x)}{n} + \frac{x(1-x)(1-2x)}{n^2}.$$

Uniform bound. On $[0,1]$, $0 \le x^2(1-x) \le \frac{4}{27}$ and $|x(1-x)(1-2x)| \le \frac{1}{4}$. Therefore

$$\sup_{x\in[0,1]} |B_n(x^3)(x) - x^3| \le \frac{4}{9n} + \frac{1}{4n^2} \xrightarrow[n\to\infty]{} 0.$$

Thus $B_n f \to f$ uniformly on $[0,1]$ for $f(x) = x^3$. $\square$

Direct computation from the Bernstein formula (no expectation). For $t \in [0,1]$,

$$B_n(x^3)(t) = \sum_{k=0}^{n} \left(\frac{k}{n}\right)^3 \binom{n}{k} t^k (1-t)^{n-k}.$$

Using $k^3 = (k)_3 + 3(k)_2 + (k)_1$ and the identities

$$k\binom{n}{k} = n\binom{n-1}{k-1}, \quad k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}, \quad k(k-1)(k-2)\binom{n}{k} = n(n-1)(n-2)\binom{n-3}{k-3},$$

we obtain

$$\sum_{k=0}^{n} k^3 \binom{n}{k} t^k (1-t)^{n-k} = n(n-1)(n-2)\, t^3 + 3n(n-1)\, t^2 + n\, t.$$

Dividing by n^3 yields

$$B_n(x^3)(t) = t^3 + \frac{3t^2 - 3t^3}{n} + \frac{t - 3t^2 + 2t^3}{n^2},$$

so

$$B_n(x^3)(t) - t^3 = \frac{3t^2(1-t)}{n} + \frac{t(1-t)(1-2t)}{n^2}.$$

Since $0 \le t^2(1-t) \le \frac{4}{27}$ and $|t(1-t)(1-2t)| \le \frac{1}{4}$ on $[0,1]$,

$$\sup_{t\in[0,1]} |B_n(x^3)(t) - t^3| \le \frac{4}{9n} + \frac{1}{4n^2} \to 0,$$

hence $B_n(x^3) \to x^3$ uniformly on $[0,1]$.

Chebyshev polynomials (first kind)

They are defined by $T_n(\cos\theta) = \cos(n\theta)$ and satisfy $T_0(x) = 1$, $T_1(x) = x$, and $T_{n+1}(x) = 2x\, T_n(x) - T_{n-1}(x)$.

The first five are

$$\begin{aligned} T_0(x) &= 1, \\ T_1(x) &= x, \\ T_2(x) &= 2x^2 - 1, \\ T_3(x) &= 4x^3 - 3x, \\ T_4(x) &= 8x^4 - 8x^2 + 1. \end{aligned}$$

(Optionally, $T_5(x) = 16x^5 - 20x^3 + 5x$.)

Legendre polynomials: Gram–Schmidt, endpoint vanishing, and Rodrigues-type P_n

Inner product. $\langle f, g\rangle = \displaystyle\int_{-1}^{1} f(x)g(x)\, dx.$

(i) First four Legendre polynomials via Gram–Schmidt

$$p_0(x) = 1, \qquad p_1(x) = x, \qquad p_2(x) = \tfrac{1}{2}(3x^2 - 1), \qquad p_3(x) = \tfrac{1}{2}(5x^3 - 3x).$$

(Outline: orthogonalize $1, x, x^2, x^3$. Use $\langle x, 1\rangle = 0$, $x^2 - (\langle x^2, 1\rangle/\langle 1, 1\rangle) = x^2 - \frac{1}{3}$, and $x^3 - (\langle x^3, x\rangle/\langle x, x\rangle)x = x^3 - \frac{3}{5}x$, then scale to the standard normalization $p_n(1) = 1$.)

(ii) Endpoint vanishing and orthogonality

For $\phi_n(x) = (1-x)^n(1+x)^n = (1-x^2)^n$, $x = \pm 1$ is a zero of multiplicity n, hence

$$\frac{d^m}{dx^m}\phi_n(\pm 1) = 0 \quad \text{for all } m < n.$$

Define $P_n(x) := \dfrac{d^n}{dx^n}(1-x^2)^n$. For $m < n$,

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = \int \phi_n^{(n)}\,\phi_m^{(m)} = (-1)^n \int \phi_n\,\phi_m^{(m+n)} = 0,$$

where we integrated by parts n times; all boundary terms vanish by the endpoint-vanishing above, and $\phi_m^{(m+n)} \equiv 0$ since $\deg \phi_m = 2m < m+n$. Hence the P_n form an orthogonal family of degree n and therefore are scalar multiples of the Legendre polynomials p_n.

(iii) Computing P_n for $n = 0, 1, 2, 3$

$$\begin{aligned}
P_0(x) &= 1,\\
P_1(x) &= -2x = (-2)\,p_1(x),\\
P_2(x) &= -4 + 12x^2 = 8\,p_2(x),\\
P_3(x) &= 72x - 120x^3 = (-48)\,p_3(x).
\end{aligned}$$

Thus $P_n = (-1)^n 2^n n!\,p_n$ for $n = 0, 1, 2, 3$, confirming the conclusion in (ii).

Moments determine a continuous function

Claim. If $f \in C[0,1]$ and $\int_0^1 f(x)x^n\,dx = 0$ for all $n = 0, 1, 2, \ldots$, then $f \equiv 0$. If instead $\int_0^1 f(x)x^n\,dx = 0$ holds for all $n \geq 1$, the same conclusion still holds.

Proof (case $n \geq 0$). Define $L : C[0,1] \to \mathbb{R}$ by $L(g) = \int_0^1 f(x)g(x)\,dx$. Then L is continuous and $L(p) = 0$ for every polynomial p. By the Weierstrass theorem, polynomials are dense in $C[0,1]$, hence $L \equiv 0$. Taking $g = f$ gives $\int_0^1 f(x)^2\,dx = 0$, so $f \equiv 0$.

Proof (case $n \geq 1$ only). Again let $L(g) = \int_0^1 f(x)g(x)\,dx$. If p is a polynomial with $p(0) = 0$, then $p(x) = \sum_{k=1}^m a_k x^k$, so $L(p) = 0$ by hypothesis. By density of such polynomials in $\{g \in C[0,1] : g(0) = 0\}$, we obtain $L(g) = 0$ for every continuous g with $g(0) = 0$. Therefore, for arbitrary $\varphi \in C[0,1]$,

$$L(\varphi) = L\big(\varphi - \varphi(0)\big) + \varphi(0)L(1) = c\,\varphi(0), \qquad c := \int_0^1 f.$$

Let φ_ε be a continuous function with $\varphi_\varepsilon(0) = 1$ and $\operatorname{supp}\varphi_\varepsilon \subset [0, \varepsilon]$. Then

$$c = L(\varphi_\varepsilon) = \int_0^1 f(x)\varphi_\varepsilon(x)\,dx \quad \text{and} \quad |L(\varphi_\varepsilon)| \leq \|f\|_\infty\,\varepsilon \to 0,$$

hence $c = 0$. Thus $L(\varphi) = 0$ for all $\varphi \in C[0,1]$, and taking $\varphi = f$ yields $\int_0^1 f(x)^2\,dx = 0$. By continuity, $f \equiv 0$. $\square$

Remark. Without continuity, the second statement is false: the point mass δ_0 has $\int_0^1 x^n\,d\delta_0 = 0$ for all $n \geq 1$ but is not the zero function.

Orthogonalization of $1, x, x^2, \ldots$ on $[-1, 1]$ (Legendre via Gram–Schmidt)

Inner product. $\langle f, g \rangle = \displaystyle\int_{-1}^{1} f(x)g(x)\,dx$. Useful integrals: $\int_{-1}^{1} x^m\,dx = 0$ for m odd, and $= 2/(m+1)$ for m even.

Results (scaled so $p_n(1) = 1$).

$$\begin{aligned}
p_0(x) &= 1, \\
p_1(x) &= x, \\
p_2(x) &= \tfrac{1}{2}(3x^2 - 1), \\
p_3(x) &= \tfrac{1}{2}(5x^3 - 3x), \\
p_4(x) &= \tfrac{1}{8}(35x^4 - 30x^2 + 3).
\end{aligned}$$

Sketch of Gram–Schmidt steps.

$$\begin{aligned}
u_0 &= 1 \;\Rightarrow\; p_0 = 1, \\
u_1 &= x - \frac{\langle x, 1\rangle}{\langle 1, 1\rangle} 1 = x \;\Rightarrow\; p_1 = x, \\
u_2 &= x^2 - \frac{\langle x^2, 1\rangle}{\langle 1, 1\rangle} 1 = x^2 - \tfrac{1}{3} \;\Rightarrow\; p_2 = \tfrac{3}{2}\big(x^2 - \tfrac{1}{3}\big), \\
u_3 &= x^3 - \frac{\langle x^3, x\rangle}{\langle x, x\rangle} x = x^3 - \tfrac{3}{5}x \;\Rightarrow\; p_3 = \tfrac{5}{2}\big(x^3 - \tfrac{3}{5}x\big), \\
u_4 &= x^4 - \frac{\langle x^4, 1\rangle}{\langle 1, 1\rangle} 1 - \frac{\langle x^4, p_2\rangle}{\langle p_2, p_2\rangle} p_2 = x^4 - \tfrac{6}{7}x^2 + \tfrac{3}{35}, \\
p_4 &= \frac{35}{8}\,u_4 = \tfrac{1}{8}(35x^4 - 30x^2 + 3).
\end{aligned}$$

Orthonormal scaling. $\displaystyle\int_{-1}^{1} p_n^2 = \frac{2}{2n+1}$, so $\phi_n(x) = \sqrt{\dfrac{2n+1}{2}}\,p_n(x)$ is orthonormal.

Orthogonality of Chebyshev polynomials and the weight

Let T_n be the Chebyshev polynomials of the first kind, defined by $T_n(\cos\theta) = \cos(n\theta)$. Set

$$w(x) = \frac{1}{\sqrt{1 - x^2}}, \qquad x \in (-1, 1).$$

Then for all $m, n \geq 0$,

$$\int_{-1}^{1} T_m(x)\,T_n(x)\,w(x)\,dx = \int_{-1}^{1} T_m(x)T_n(x)\frac{dx}{\sqrt{1-x^2}} = \int_{0}^{\pi} \cos(m\theta)\cos(n\theta)\,d\theta,$$

where we used the substitution $x = \cos\theta$ (so $dx = -\sin\theta\,d\theta$ and $dx/\sqrt{1-x^2} = -d\theta$). By the standard cosine orthogonality on $[0, \pi]$,

$$\int_{0}^{\pi} \cos(m\theta)\cos(n\theta)\,d\theta = \begin{cases} 0, & m \neq n, \\ \pi, & m = n = 0, \\ \pi/2, & m = n \geq 1. \end{cases}$$

Therefore $\{T_n\}_{n\geq 0}$ is an orthogonal system on $[-1,1]$ with respect to $w(x) = (1-x^2)^{-1/2}$. An orthonormal system is given by

$$\phi_0(x) = \frac{1}{\sqrt{\pi}}, \qquad \phi_n(x) = \sqrt{\frac{2}{\pi}}\, T_n(x) \quad (n \geq 1).$$

Quadrature with nodes and weights; exactness and Gauss rules

Fix an interval $[a,b]$. For $n \in \mathbb{N}$, let nodes $x_k^{(n)} \in [a,b]$ and weights $A_k^{(n)} \geq 0$ be given $(k = 1, \dots, n)$. Define

$$Q_n(P) := \sum_{k=1}^{n} A_k^{(n)}\, P\big(x_k^{(n)}\big), \qquad \varepsilon_n(P) := \left| \int_a^b P(x)\,dx \;-\; Q_n(P) \right|.$$

Exactness and moment conditions. The rule is exact for all polynomials of degree $\leq m$ if and only if

$$\sum_{k=1}^{n} A_k^{(n)} \big(x_k^{(n)}\big)^j = \int_a^b x^j\,dx \qquad (j = 0, 1, \dots, m),$$

i.e. the first $m+1$ moments are matched.

Sharp upper bound. No n-node rule can be exact beyond degree $2n-1$. Indeed, if it were exact for degree $2n$, then with $\pi(x) = \prod_{k=1}^{n}(x - x_k^{(n)})^2 \geq 0$ we would get $0 = Q_n(\pi) = \sum_k A_k^{(n)} \pi(x_k^{(n)}) = \int_a^b \pi(x)\,dx > 0$, a contradiction.

Gauss–Legendre rule (achieves $2n-1$). Let P_n be the degree-n Legendre polynomial on $[-1,1]$ and let $\xi_1, \dots, \xi_n$ be its zeros. Set nodes on $[a,b]$ by $x_k^{(n)} = \frac{a+b}{2} + \frac{b-a}{2}\,\xi_k$ and weights by

$$A_k^{(n)} = \frac{b-a}{2}\,\omega_k, \qquad \omega_k = \frac{2}{\big(1-\xi_k^2\big)\,[P_n'(\xi_k)]^2}\ (> 0).$$

Then

$$\sum_{k=1}^{n} A_k^{(n)}\, P\big(x_k^{(n)}\big) = \int_a^b P(x)\,dx \qquad \text{for all polynomials } \deg P \leq 2n-1.$$

Sketch: write $p = qP_n + r$ with $\deg q, \deg r \leq n-1$; orthogonality gives $\int qP_n = 0$ and $P_n(\xi_k) = 0$ gives $p(\xi_k) = r(\xi_k)$; the quadrature integrates r exactly.

Examples on $[-1,1]$.

- $n = 1$: node 0, weight 2; exact for $\deg \leq 1$.
- $n = 2$: nodes $\pm 1/\sqrt{3}$, weights $1, 1$; exact for $\deg \leq 3$.
- $n = 3$: nodes $0,\ \pm\sqrt{3/5}$; weights $8/9,\ 5/9,\ 5/9$; exact for $\deg \leq 5$.

Mapping to $[a,b]$ uses $x = \frac{a+b}{2} + \frac{b-a}{2}\xi$ and multiplies all weights by $\frac{b-a}{2}$.

Function Approximation vs. Quadrature (on $[-1,1]$)

A. Polynomial approximation of a continuous function

Goal. Given $f \in C([-1,1])$, choose a polynomial p_n so that

$$\|f - p_n\|_\infty = \sup_{x\in[-1,1]} |f(x) - p_n(x)|$$

is small (ideally, minimal). This targets the *entire function* on the whole interval.

Tools. Weierstrass approximation (existence of good polynomials), Chebyshev (minimax/near-best choices), Jackson-type estimates (rates in terms of smoothness).

B. Numerical quadrature (this task)

Goal. Approximate the *single number*

$$I(f) = \int_{-1}^{1} f(x)\,dx$$

by a weighted sum of *finitely many samples*:

$$Q_n(f) = \sum_{k=1}^{n} A_k\, f(x_k), \qquad x_k \in [-1,1],\ A_k > 0.$$

We design x_k, A_k so that Q_n is *exact* for all polynomials of degree $\le m$:

$$Q_n(p) = \int_{-1}^{1} p(x)\,dx \quad \text{for all } \deg p \le m.$$

Any n-node rule has $m \le 2n-1$ (sharp). *Gauss–Legendre* quadrature (zeros of the Legendre polynomial P_n as nodes, specific positive weights) achieves $m = 2n-1$.

C. Same actors, different roles

Orthogonal polynomials appear in both settings:

- In *approximation*, expanding f in Chebyshev/Legendre series and truncating gives small $\|f - p_n\|_\infty$.
- In *quadrature*, Gauss nodes are the zeros of an orthogonal polynomial, yielding exactness up to degree $2n-1$.

D. Tiny example

For $f(x) = x^4$ on $[-1,1]$:

- A degree-1 polynomial $p(x) = ax + b$ cannot uniformly approximate x^4 well; the optimal error is $O(1)$.
- The 3-point Gauss rule (degree of exactness 5) integrates x^4 *exactly* using just 3 samples, even though it does not approximate the whole function.

E. Side-by-side comparison

Aspect	**Approximation**	**Quadrature**
Target	$f(\cdot)$ on $[-1,1]$	$\int_{-1}^{1} f$ (a scalar)
Output	Polynomial p_n	Nodes x_k, weights A_k
Error	$\|f - p_n\|_\infty$ (uniform)	$\left\lvert \int f - \sum A_k f(x_k) \right\rvert$
Optimality	Minimax/Chebyshev, etc.	Degree of exactness $m \le 2n-1$
Data used	Potentially dense info on f	Only n samples $f(x_k)$
Key tools	Weierstrass, Chebyshev series	Orthogonal polys, Gauss nodes

No n-node quadrature can be exact beyond degree $2n-1$

Let $\alpha_1, \dots, \alpha_n \in [-1,1]$ be distinct nodes and $A_1, \dots, A_n \in \mathbb{R}$ weights. Consider the quadrature $Q_n(p) = \sum_{j=1}^n A_j\, p(\alpha_j)$.

Theorem 4. *For any choice of nodes and weights, the degree of exactness of Q_n is at most $2n-1$. In particular, there do not exist n distinct nodes and real weights such that*

$$\int_{-1}^{1} p(x)\,dx = \sum_{j=1}^{n} A_j\, p(\alpha_j) \quad \textit{for all } \deg p \le 2n.$$

Proof. Define $\pi(x) = \prod_{j=1}^{n} (x-\alpha_j)^2$. Then $\deg \pi = 2n$, and $\pi(\alpha_j) = 0$ for all j, so $Q_n(\pi) = 0$. Since $\pi \ge 0$ and $\pi \not\equiv 0$, we have $\int_{-1}^{1} \pi(x)\,dx > 0$. If the rule were exact for degree $2n$, we would have $Q_n(\pi) = \int_{-1}^{1} \pi$, a contradiction. Hence the degree of exactness is $\le 2n-1$. $\square$

Optimality. The Gauss–Legendre rule, with nodes the zeros of the Legendre polynomial P_n and the usual positive weights, is exact for all $\deg p \le 2n-1$, so the bound is sharp.

Checks. For $n = 1$, exactness for $1, x, x^2$ forces $A_1 = 2$, $\alpha_1 = 0$, and $2\alpha_1^2 = 2/3$, impossible. For $n = 2$ with symmetric nodes $\pm c$ and equal weights w, exactness for $1, x^2, x^4$ yields $w = 1$, $c^2 = 1/3$, and $c^4 = 1/5$, a contradiction.

What an n-point quadrature rule means

Definition. An n-point quadrature rule on $[a,b]$ is specified by *nodes* $x_1, \dots, x_n \in [a,b]$ and *weights* $A_1, \dots, A_n$ and approximates

$$\int_a^b f(x)\,dx \;\approx\; Q_n(f) := \sum_{k=1}^{n} A_k\, f(x_k).$$

Thus the rule requires exactly n function evaluations $f(x_k)$.

Degree of exactness. The rule is *exact of degree m* if $Q_n(p) = \int_a^b p(x)\,dx$ for every polynomial p with $\deg p \le m$. With only function values (no derivatives), any n-point rule satisfies $m \le 2n-1$ (sharp). Gauss–Legendre quadrature attains $m = 2n-1$.

Closed vs. open rules. Nodes may include endpoints (*closed* rules, e.g. trapezoid, Simpson) or avoid them (*open* rules, e.g. midpoint). In either case there are n nodes and n evaluations.

Composite/adaptive use. To integrate on $[a,b]$ one often partitions the interval and applies the n-point rule on each subinterval; the overall method then uses many points (roughly n times the number of panels), but per panel it remains an n-point quadrature.

Examples on $[-1,1]$.

- 2-point Gauss: nodes $\pm 1/\sqrt{3}$, weights $1, 1$; exact for $\deg \le 3$.
- Simpson's rule (3 points): nodes $-1, 0, 1$, weights $\frac{1}{3}(1,4,1)$; exact for $\deg \le 3$.

Examples of n-point quadrature rules and what you evaluate

Definition. An n-point rule on $[a,b]$ has nodes $x_1,\dots,x_n \in [a,b]$ and weights $A_1,\dots,A_n$ and computes

$$\int_a^b f(x)\,dx \approx Q_n(f) := \sum_{k=1}^{n} A_k\, f(x_k),$$

so you evaluate f at exactly n points x_k.

Standard rules (nodes, weights, degree of exactness). On $[a,b]$ we have:

Rule	Nodes x_k	Weights A_k	Exact for $\deg \le$
Midpoint ($n=1$)	$\frac{a+b}{2}$	$b-a$	1
Trapezoid ($n=2$)	$a,\ b$	$\frac{b-a}{2},\ \frac{b-a}{2}$	1
Simpson ($n=3$)	$a,\ \frac{a+b}{2},\ b$	$\frac{b-a}{6}\cdot(1,4,1)$	3
Gauss–Legendre ($n=2$)	$\frac{a+b}{2} \pm \frac{b-a}{2\sqrt{3}}$	$\frac{b-a}{2},\ \frac{b-a}{2}$	3
Gauss–Legendre ($n=3$)	$\frac{a+b}{2},\ \frac{a+b}{2} \pm \frac{b-a}{2}\sqrt{\frac{3}{5}}$	$\frac{b-a}{2}\cdot\left(\frac{8}{9},\frac{5}{9},\frac{5}{9}\right)$	5

What you actually compute. For example, with the 3-point Gauss rule on $[a,b]$:

$$Q_3(f) = \frac{b-a}{2}\left[\frac{8}{9}\,f\!\left(\tfrac{a+b}{2}\right) + \frac{5}{9}\,f\!\left(\tfrac{a+b}{2} - \tfrac{b-a}{2}\sqrt{\tfrac{3}{5}}\right) + \frac{5}{9}\,f\!\left(\tfrac{a+b}{2} + \tfrac{b-a}{2}\sqrt{\tfrac{3}{5}}\right)\right],$$

which uses *exactly three* evaluations of f.

Sanity check on $[-1,1]$ for $f(x)=x^4$ (true integral $=2/5$).

$$\begin{aligned}
\text{Simpson: } & \frac{2}{6}\,[f(-1)+4f(0)+f(1)] = \frac{1}{3}(1+0+1) = \frac{2}{3},\\
\text{Gauss 2-point: } & f(-1/\sqrt{3}) + f(1/\sqrt{3}) = 2\cdot\left(\frac{1}{\sqrt{3}}\right)^4 = \frac{2}{9},\\
\text{Gauss 3-point: } & \frac{8}{9}\,f(0) + \frac{5}{9}\,f(\pm\sqrt{3/5}) = \frac{10}{9}\left(\frac{3}{5}\right)^2 = \frac{2}{5}.
\end{aligned}$$

Only the 3-point Gauss rule (degree 5) integrates the quartic exactly.

Approximating integrals by exactness on polynomials; Legendre background

Quadrature and degree of exactness. An n-point rule on $[-1,1]$ has nodes $x_1,\dots,x_n$ and weights $A_1,\dots,A_n$ and computes

$$Q_n(f) = \sum_{k=1}^{n} A_k f(x_k).$$

It is *exact of degree m* if $Q_n(p) = \int_{-1}^{1} p(x)\,dx$ for every polynomial p with $\deg p \le m$. With n nodes (no derivatives) one always has $m \le 2n-1$ (sharp).

Legendre polynomials. They form an orthogonal basis on $[-1,1]$ (weight 1):

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = \begin{cases} \dfrac{2}{2n+1}, & m=n,\\ 0, & m\ne n.\end{cases}$$

Rodrigues: $P_n(x) = \dfrac{1}{2^n n!}\dfrac{d^n}{dx^n}(x^2-1)^n$. Recurrence: $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$.

Gauss–Legendre quadrature (achieves $2n-1$). Let $\xi_1, \dots, \xi_n$ be the zeros of P_n. Set

$$\omega_k = \frac{2}{(1-\xi_k^2)\,[P_n'(\xi_k)]^2} \quad (\text{all } \omega_k > 0), \qquad Q_n(f) = \sum_{k=1}^{n} \omega_k\, f(\xi_k).$$

Then $Q_n(p) = \int_{-1}^{1} p(x)\,dx$ for every $\deg p \le 2n-1$. *Sketch:* write $p = qP_n + r$ with $\deg q, \deg r \le n-1$; use $\int qP_n = 0$ and $P_n(\xi_k) = 0$, and the fact that Q_n integrates any degree-$\le n-1$ polynomial exactly.

Examples on $[-1,1]$.

- $n = 2$: nodes $\pm 1/\sqrt{3}$, weights $1, 1$; exact for $\deg \le 3$.
- $n = 3$: nodes $0,\ \pm\sqrt{3/5}$, weights $8/9,\ 5/9,\ 5/9$; exact for $\deg \le 5$.

Impossibility beyond $2n-1$. For any nodes $\{\alpha_j\}$, the polynomial $\pi(x) = \prod_{j=1}^{n}(x-\alpha_j)^2$ has degree $2n$, satisfies $Q_n(\pi) = 0$, but $\int_{-1}^{1} \pi(x)\,dx > 0$. Hence no n-point rule can be exact for all $\deg 2n$ polynomials.

Building a 10-point quadrature for $[a, b]$

Goal. Approximate $I(f) = \displaystyle\int_a^b f(x)\,dx$ from $n = 10$ samples of f.

A. Choosing the nodes (Gauss–Legendre, optimal)

Let (ξ_i, ω_i) be the 10-point Gauss–Legendre nodes/weights on $[-1,1]$:

$$\begin{aligned}
\xi_{1,10} &= \pm 0.9739065285171717, & \omega_{1,10} &= 0.0666713443086881,\\
\xi_{2,9} &= \pm 0.8650633666889845, & \omega_{2,9} &= 0.1494513491505806,\\
\xi_{3,8} &= \pm 0.6794095682990244, & \omega_{3,8} &= 0.2190863625159820,\\
\xi_{4,7} &= \pm 0.4333953941292472, & \omega_{4,7} &= 0.2692667193099963,\\
\xi_{5,6} &= \pm 0.1488743389816312, & \omega_{5,6} &= 0.2955242247147529.
\end{aligned}$$

Map to $[a, b]$ by

$$x_i = \frac{a+b}{2} + \frac{b-a}{2}\,\xi_i, \qquad A_i = \frac{b-a}{2}\,\omega_i,$$

and compute

$$Q_{10}(f) = \sum_{i=1}^{10} A_i\, f(x_i).$$

This rule is exact for all polynomials with $\deg \le 19$.

B. Nodes are given (interpolatory quadrature)

Let distinct $x_1, \dots, x_{10} \in [a, b]$ be given and define the Lagrange basis

$$\ell_k(x) = \prod_{\substack{j=1\\ j\ne k}}^{10} \frac{x - x_j}{x_k - x_j}.$$

Set the weights by

$$A_k = \int_a^b \ell_k(x)\,dx, \qquad k = 1, \dots, 10.$$

Then

$$Q_{10}(f) = \sum_{k=1}^{10} A_k\, f(x_k)$$

is exact for all polynomials of degree ≤ 9. Equivalently, $\{A_k\}$ solve the moment system

$$\sum_{k=1}^{10} A_k\, x_k^j = \frac{b^{j+1} - a^{j+1}}{j+1}, \qquad j = 0, 1, \ldots, 9.$$

Remarks. (1) Affine-scaling to $[-1, 1]$ before forming ℓ_k or the moment system improves conditioning.
(2) If the nodes are equally spaced (including endpoints), this yields the closed Newton–Cotes rule with 10 points, which is less stable than Gauss–Legendre.

Chebyshev tail with nonnegative coefficients: equioscillation of the truncation error

Let T_j be the Chebyshev polynomials of the first kind and let $\{\gamma_j\}_{j\ge 1}$ be nonnegative with $\sum_{j=1}^{\infty} \gamma_j < \infty$. Define

$$f(x) = \sum_{j=1}^{\infty} \gamma_j\, T_{3j}(x), \qquad P_n(x) = \sum_{j=1}^{n} \gamma_j\, T_{3j}(x), \qquad R_n(x) = f(x) - P_n(x) = \sum_{j=n+1}^{\infty} \gamma_j\, T_{3j}(x).$$

(1) Continuity. Since $|T_{3j}(x)| \le 1$ on $[-1, 1]$ and $\sum \gamma_j < \infty$, the series for f converges absolutely and uniformly (M–test), so $f \in C([-1, 1])$.

(2) Exact size of the tail. For all $x \in [-1, 1]$, $|R_n(x)| \le \sum_{j=n+1}^{\infty} \gamma_j =: E_n$. At $x = 1$, $T_{3j}(1) = 1$, hence $|R_n(1)| = E_n$. Therefore $\|f - P_n\|_\infty = \|R_n\|_\infty = E_n$.

(3) Minimax property of P_n. Let $\mathcal{P}_{3n}$ be the polynomials of degree $\le 3n$. For any $Q \in \mathcal{P}_{3n}$, write $Q = \sum_{k=0}^{3n} a_k T_k$. Then

$$f - Q = \sum_{j=1}^{n} (\gamma_j - a_{3j}) T_{3j} \;+\; \sum_{\substack{0 \le k \le 3n \\ k \not\equiv 0\,(3)}} (-a_k) T_k \;+\; R_n.$$

Using $\left\|\sum c_\ell T_\ell\right\|_\infty \le \sum |c_\ell|$ gives

$$\|f - Q\|_\infty \;\ge\; \|R_n\|_\infty - \sum_{j=1}^{n} \bigl|\gamma_j - a_{3j}\bigr| - \sum_{k \not\equiv 0(3)} |a_k| \;\ge\; \|R_n\|_\infty = E_n,$$

with equality iff $a_{3j} = \gamma_j$ for $1 \le j \le n$ and $a_k = 0$ for $k \not\equiv 0 \pmod 3$, i.e. iff $Q = P_n$. Thus P_n is the minimax polynomial in $\mathcal{P}_{3n}$ and the best error equals E_n.

(4) Equioscillation points. By the Chebyshev alternation theorem for best uniform approximation by $\mathcal{P}_{3n}$, there exist points

$$-1 \le x_0 < x_1 < \cdots < x_{3n+1} \le 1$$

such that

$$f(x_k) - P_n(x_k) = (-1)^k\, \|f - P_n\|_\infty = (-1)^k \sum_{j=n+1}^{\infty} \gamma_j, \qquad k = 0, 1, \ldots, 3n+1.$$

Using 10 points to approximate $\int_a^b f(x)\,dx$

Gauss–Legendre (you may choose the points). Let $(\xi_i, \omega_i)_{i=1}^{10}$ be the 10-point Gauss–Legendre nodes/weights on $[-1,1]$. Map to $[a,b]$ by

$$x_i = \frac{a+b}{2} + \frac{b-a}{2}\,\xi_i, \qquad A_i = \frac{b-a}{2}\,\omega_i.$$

Then the quadrature

$$Q_{10}(f) = \sum_{i=1}^{10} A_i\, f(x_i)$$

is exact for every polynomial p with $\deg p \le 19$ (optimal among all 10-point rules), and all $A_i > 0$.

Interpolatory rule (points are pre-given). If distinct $x_1, \dots, x_{10} \in [a,b]$ are given, define the Lagrange basis

$$\ell_k(x) = \prod_{\substack{j=1\\ j\neq k}}^{10} \frac{x - x_j}{x_k - x_j}, \qquad A_k = \int_a^b \ell_k(x)\,dx.$$

Then

$$Q_{10}(f) = \sum_{k=1}^{10} A_k\, f(x_k)$$

is exact for all polynomials with deg ≤ 9. (Equivalently, the weights solve $\sum_{k=1}^{10} A_k x_k^j = (b^{j+1} - a^{j+1})/(j+1)$ for $j = 0, \dots, 9$.)

Problem 4

For each $f \in C([-1,1])$, let $\mathcal{P}_n$ denote the subspace of polynomials of degree at most n, and define the best uniform approximation error

$$E_n(f) := \inf_{p\in\mathcal{P}_n} \sup_{x\in[-1,1]} |f(x) - p(x)|.$$

By the Weierstrass approximation theorem, $E_n(f) \to 0$ for every $f \in C([-1,1])$. Using the result of Question 3, construct a function $f \in C([-1,1])$ to show that the convergence $E_n(f) \to 0$ can be *arbitrarily slow* in the following sense:

Claim to prove. For any decreasing sequence $(\delta_n)_{n\ge1}$ of nonnegative numbers with $\delta_n \to 0$, there exists $f \in C([-1,1])$ such that

$$E_n(f) \;\ge\; \delta_n \qquad \text{for all } n = 1, 2, \dots.$$

Arbitrarily slow uniform polynomial approximation

For $f \in C([-1,1])$ let

$$E_n(f) := \inf_{p\in\mathcal{P}_n} \|f - p\|_\infty, \qquad \mathcal{P}_n = \{p : \deg p \le n\}.$$

By Weierstrass, $E_n(f) \downarrow 0$. We show the convergence can be arbitrarily slow.

Lemma (from Problem 3). If $\gamma_j \ge 0$ with $\sum_{j=1}^{\infty} \gamma_j < \infty$ and

$$f(x) = \sum_{j=1}^{\infty} \gamma_j\, T_{3j}(x), \qquad P_n(x) = \sum_{j=1}^{n} \gamma_j\, T_{3j}(x),$$

then $f \in C([-1,1])$, P_n is the best uniform approximant of degree $\le 3n$, and

$$E_{3n}(f) = \|f - P_n\|_\infty = \sum_{j=n+1}^{\infty} \gamma_j.$$

Theorem (arbitrarily slow decay of $E_n(f)$). Given any decreasing sequence $\delta_n \ge 0$ with $\delta_n \to 0$, there exists $f \in C([-1,1])$ such that $E_n(f) \ge \delta_n$ for all $n \ge 1$.

Proof. Set $s_n := \delta_{3n}$ and define $\gamma_{n+1} := s_n - s_{n+1} \ge 0$. Then $\sum \gamma_j = s_0 - \lim s_n < \infty$ and hence $f(x) := \sum_{j=1}^{\infty} \gamma_j T_{3j}(x) \in C([-1,1])$. By the lemma,

$$E_{3n}(f) = \sum_{j=n+1}^{\infty} \gamma_j = s_n = \delta_{3n}.$$

Since $E_n(f)$ is nonincreasing in n, for any $m \in \{3n, 3n+1, 3n+2\}$,

$$E_m(f) \ge E_{3n}(f) = \delta_{3n} \ge \delta_m,$$

because δ_n is decreasing. Therefore $E_n(f) \ge \delta_n$ for all n. $\square$

Examples. Choosing $\delta_n = 1/\log(n+e)$ (or $1/\log\log(n+e^e)$, etc.) yields a continuous f for which $E_n(f)$ decreases as slowly as the prescribed sequence.

Piecewise-linear (hat-function) interpolation on a uniform grid

Definition. For $n \in \mathbb{N}$ and $r \in \mathbb{Z}$, define

$$\Delta_{n,r}(x) := \max\Big\{0,\ 1 - n\big|x - \tfrac{r}{n}\big|\Big\}, \qquad x \in [-1,1].$$

Each $\Delta_{n,r}$ is the "hat" centered at r/n with height 1 and support $\big[(r-1)/n,\ (r+1)/n\big]$.

Sketch. $\Delta_{n,r}$ is triangular: it is linear on $[(r-1)/n, r/n]$ and on $[r/n, (r+1)/n]$, with

$$\Delta_{n,r}(r/n) = 1, \qquad \Delta_{n,r}((r \pm 1)/n) = 0.$$

For fixed x, at most two hats are nonzero and they satisfy $\sum_{m=-n}^{n} \Delta_{n,m}(x) = 1$.

Interpolator. Given $f : [-1,1] \to \mathbb{R}$, set

$$f_n(x) := \sum_{m=-n}^{n} f\big(\tfrac{m}{n}\big)\ \Delta_{n,m}(x).$$

Lemma 2 (Piecewise linearity and interpolation)**.** *For each n, f_n is linear on each subinterval $[r/n, (r+1)/n]$, and $f_n(r/n) = f(r/n)$ for $r = -n, \dots, n$.*

Proof. On $[r/n, (r+1)/n]$ only $\Delta_{n,r}$ and $\Delta_{n,r+1}$ are nonzero, both linear, and $\Delta_{n,r}(x) + \Delta_{n,r+1}(x) = 1$. Hence

$$f_n(x) = \Delta_{n,r}(x)\, f(r/n) + \Delta_{n,r+1}(x)\, f((r+1)/n)$$

is linear there. At $x = r/n$, $\Delta_{n,r}(r/n) = 1$ and all other hats vanish, so $f_n(r/n) = f(r/n)$. $\square$

Theorem 5 (Uniform convergence for continuous data)**.** *If $f \in C([-1,1])$, then $\|f_n - f\|_\infty \to 0$ as $n \to \infty$.*

Proof. Since f is uniformly continuous, for any $\varepsilon > 0$ choose $\delta > 0$ so that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Take n with $1/n < \delta$. For $x \in [r/n, (r+1)/n]$,

$$\begin{aligned} |f_n(x) - f(x)| &= \big|\Delta_{n,r}(x)(f(r/n) - f(x)) + \Delta_{n,r+1}(x)(f((r+1)/n) - f(x))\big| \\ &\le \Delta_{n,r}(x)\,|f(r/n) - f(x)| + \Delta_{n,r+1}(x)\,|f((r+1)/n) - f(x)| < \varepsilon, \end{aligned}$$

since both $|x - r/n|$ and $|x - (r+1)/n|$ are $\le 1/n < \delta$. Taking the supremum over x gives $\|f_n - f\|_\infty < \varepsilon$. $\square$

Rate under Lipschitz continuity (optional). If f is Lipschitz with constant L, then for all x,

$$|f_n(x) - f(x)| \le L \max\big\{|x - r/n|,\ |x - (r+1)/n|\big\} \le \frac{L}{n},$$

hence $\|f_n - f\|_\infty \le L/n$.

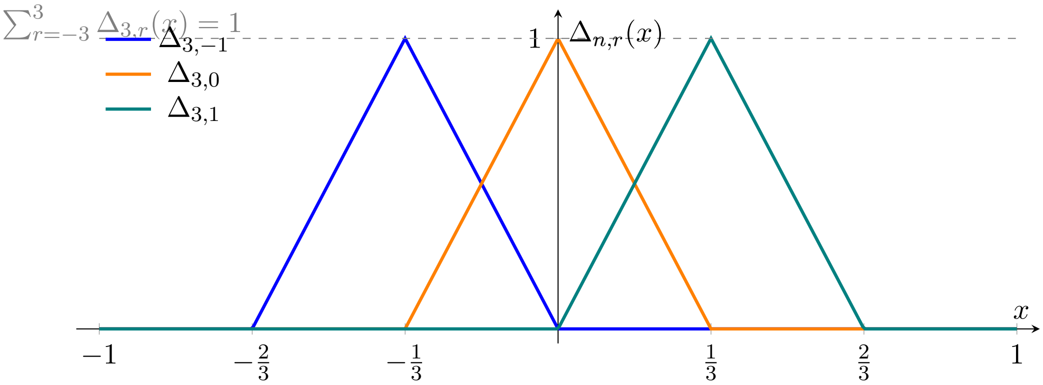


Figure 10: Three neighboring hat functions for $n = 3$. Each peak is at $x = r/n$; support width is $2/n$.

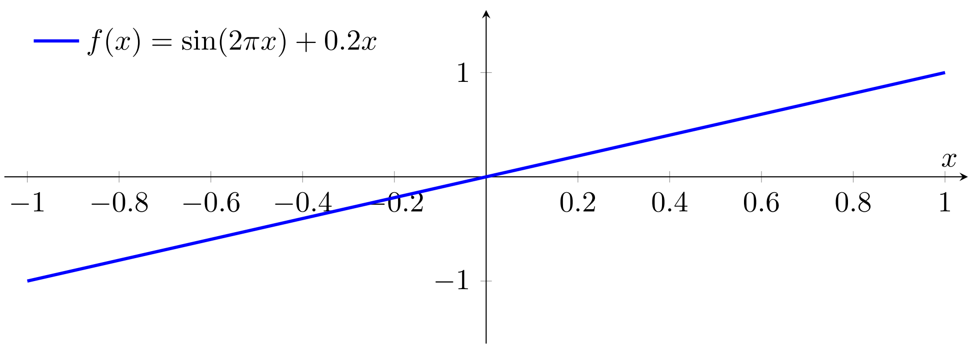


Figure 11: Piecewise-linear interpolant f_n on a uniform grid with $n = 6$.

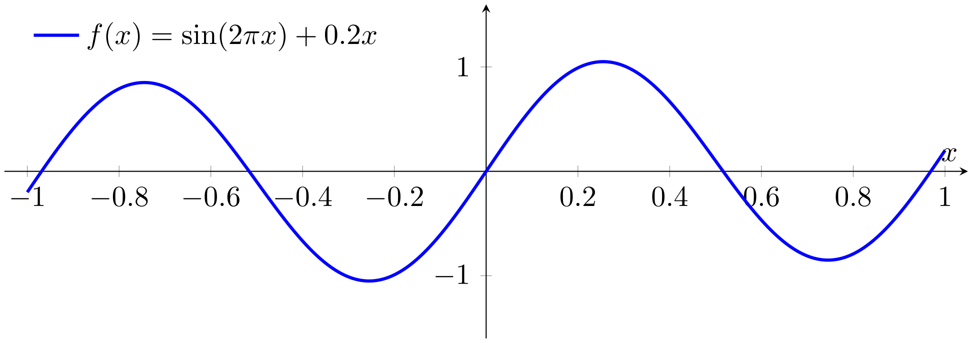


Figure 12: Target function $f(x) = \sin(2\pi x) + 0.2x$.

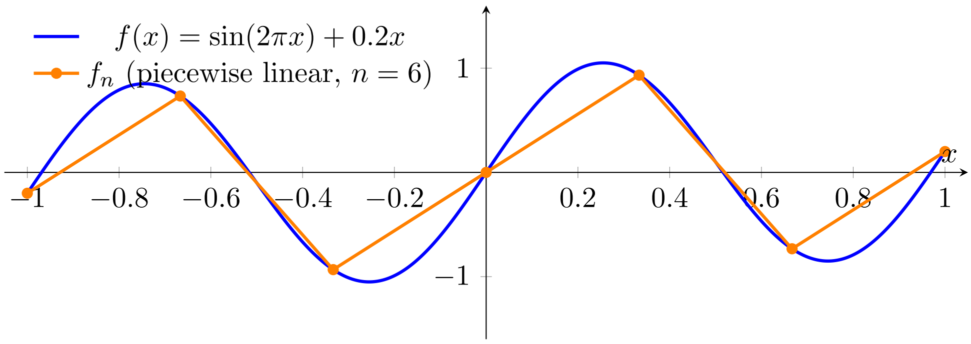


Figure 13: Target function $f(x)$ and its piecewise-linear interpolant f_n for $n = 6$.

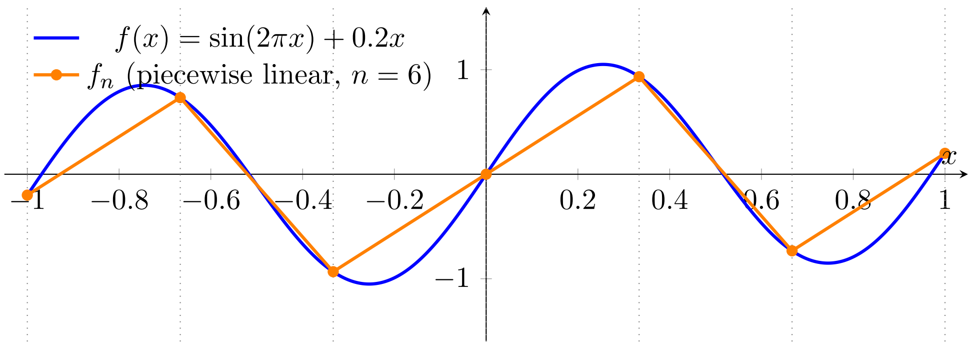


Figure 14: Target function $f(x)$ and piecewise-linear interpolant f_n for $n = 6$, with vertical grid lines at nodes.

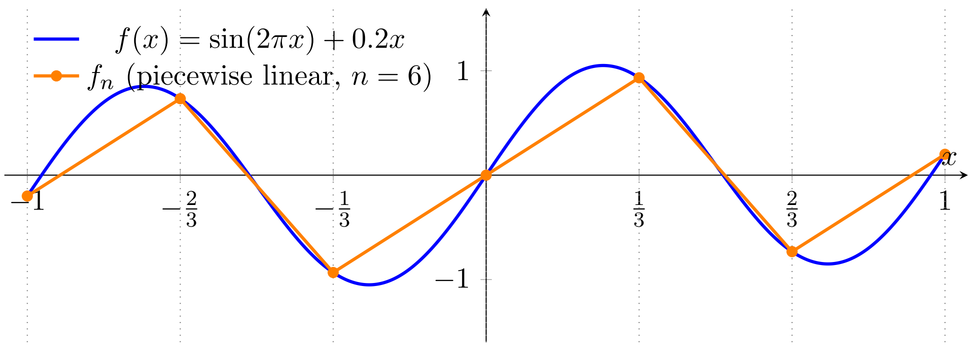


Figure 15: Comparison of $f(x)$ and piecewise-linear interpolant f_n on a uniform grid with $n = 6$, with vertical grid lines at nodes.

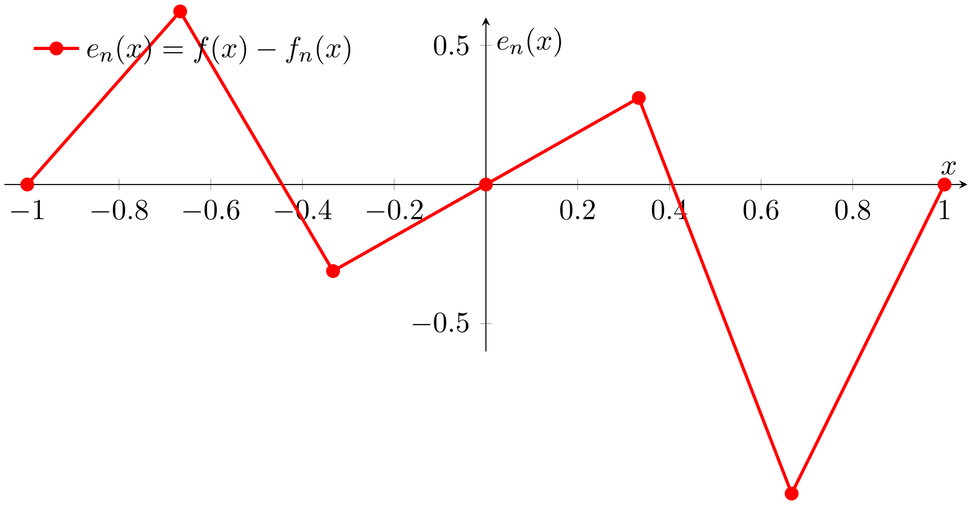


Figure 16: Interpolation error $e_n(x)$ at nodes and midpoints for $n = 6$.

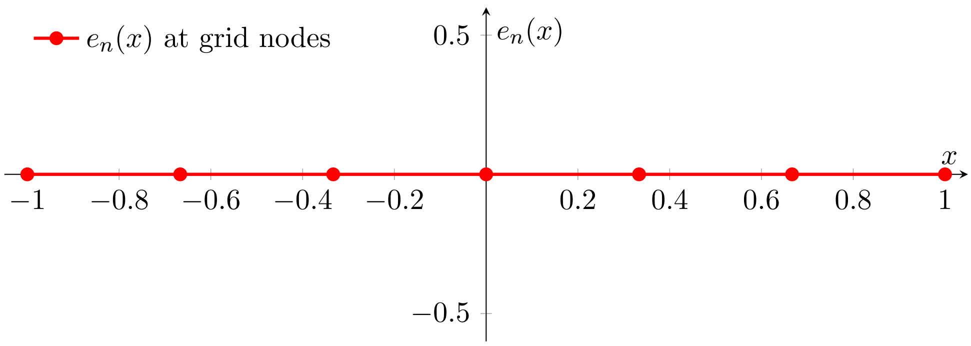


Figure 17: Interpolation error $e_n(x) = f(x) - f_n(x)$ at grid nodes ($n = 6$).

Exercise 3. Schauder basis for $C([-1,1])$. Show that there exists a sequence $(\phi_n)_{n\geq 0} \subset C([-1,1])$ such that for every $f \in C([-1,1])$ there exist unique scalars $(a_n)_{n\geq 0}$ with the series

$$\sum_{n=0}^{\infty} a_n \phi_n$$

converging uniformly to f on $[-1,1]$.

Definition 4. Schauder basis. Let X be a Banach space. A sequence $(e_n) \subset X$ is a *Schauder basis* if for every $x \in X$ there exist unique scalars (c_n) such that the partial sums $S_N = \sum_{n=0}^{N} c_n e_n$ converge to x in $\|\cdot\|_X$.

Theorem 6. *Faber–Schauder basis on* $[0,1]$. *Define $\varphi_0(t) \equiv 1$, $\varphi_1(t) = t$, and for each $m \geq 0$ and odd $k \in \{1,3,\dots,2^m-1\}$ set*

$$\varphi_{m,k}(t) := \max\Big(1 - 2^{m+1}\big|t - \tfrac{k}{2^m}\big|,\, 0\Big).$$

Enumerate the system as $\varphi_0, \varphi_1, \{\varphi_{m,k}\}_{m\geq 0,\ k \text{ odd}}$. Then (φ_n) is a Schauder basis of $C([0,1])$ with respect to $\|\cdot\|_\infty$.

Proof. Existence (uniform convergence). For $f \in C([0,1])$ define coefficients

$$a_0 := f(0), \quad a_1 := f(1) - f(0), \quad a_{m,k} := f\Big(\tfrac{k}{2^m}\Big) - \tfrac{1}{2}\Big(f\Big(\tfrac{k-1}{2^m}\Big) + f\Big(\tfrac{k+1}{2^m}\Big)\Big).$$

Let $S_M(f)$ be the finite linear combination of φ_0, φ_1 and all $\varphi_{m,k}$ with levels $\leq M$. A direct check shows $S_M(f)$ is the polygonal interpolant of f on the dyadic grid $\{j/2^M\}_{j=0}^{2^M}$, hence (for the modulus of continuity ω_f) one has

$$\|f - S_M(f)\|_\infty \leq \omega_f(2^{-M}) \xrightarrow[M\to\infty]{} 0.$$

Uniqueness. Suppose $\sum c_n \varphi_n \equiv 0$. Evaluating at $t = 0$ and $t = 1$ gives $c_0 = c_1 = 0$ (since all tents vanish at the endpoints). Next evaluate at the midpoint $t = \frac{1}{2}$: all functions from levels ≥ 1 vanish there, so $c_{0,1} = 0$. Inductively, at a new dyadic node $t = \frac{k}{2^m}$ (odd k) all higher-level tents vanish and lower levels are zero there; hence $c_{m,k} = 0$. Thus all coefficients vanish and the representation is unique. $\square$

Corollary 7. *Transfer to* $[-1,1]$. *Let $t = \frac{x+1}{2}$ be the affine bijection $[-1,1] \to [0,1]$. Define*

$$\phi_0(x) := 1, \qquad \phi_1(x) := \tfrac{x+1}{2}, \qquad \phi_{m,k}(x) := \varphi_{m,k}\Big(\tfrac{x+1}{2}\Big).$$

Then $(\phi_n) \subset C([-1,1])$ is a Schauder basis. For every $f \in C([-1,1])$ there are unique coefficients (a_n) such that $\sum_{n=0}^{\infty} a_n \phi_n$ converges uniformly to f.

Remark 5. **Explicit coefficient formulas on $[-1,1]$.** Equivalently, for $f \in C([-1,1])$ let $g(t) := f(2t-1)$ on $[0,1]$. Then

$$a_0 = g(0) = f(-1), \qquad a_1 = g(1) - g(0) = f(1) - f(-1),$$

and for $m \geq 0$, odd k,

$$a_{m,k} = g\Big(\tfrac{k}{2^m}\Big) - \tfrac{1}{2}\Big(g\Big(\tfrac{k-1}{2^m}\Big) + g\Big(\tfrac{k+1}{2^m}\Big)\Big) = f(x_{m,k}) - \tfrac{1}{2}\big(f(x_{m,k}^-) + f(x_{m,k}^+)\big),$$

where $x_{m,k} = 2 \cdot \frac{k}{2^m} - 1$ and $x_{m,k}^{\pm} = 2 \cdot \frac{k\pm 1}{2^m} - 1$.

Solution. Apply the theorem on $[0,1]$ to $g(t) := f(2t-1)$ and pull back the expansion by $t = \frac{x+1}{2}$. Uniform convergence and uniqueness are preserved by this affine change of variables. Therefore the sequence (ϕ_n) is as required.

Example 1. ***Two quick checks.***

1. *For $f(t) = t$ on $[0,1]$, $a_0 = 0$, $a_1 = 1$, and all $a_{m,k} = 0$; hence $f = \varphi_1$ exactly.*
2. *For $f(t) = t^2$ on $[0,1]$, $a_0 = 0$, $a_1 = 1$, and $a_{m,k} = \frac{k^2}{4^m} - \frac{1}{2}\frac{(k-1)^2+(k+1)^2}{4^m} = -4^{-m}$ (independent of k), so $t^2 = \varphi_1 - \sum_{m\geq 0}\sum_{k \text{ odd}} 4^{-m}\varphi_{m,k}$ with uniform convergence. Transfer to $[-1,1]$ via $t = \frac{x+1}{2}$.*

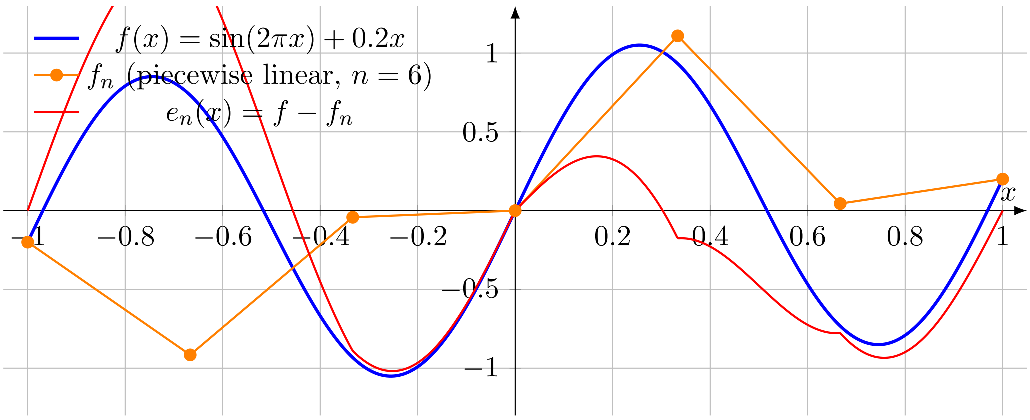


Figure 18: Function, piecewise–linear interpolant ($n = 6$), and error curve.

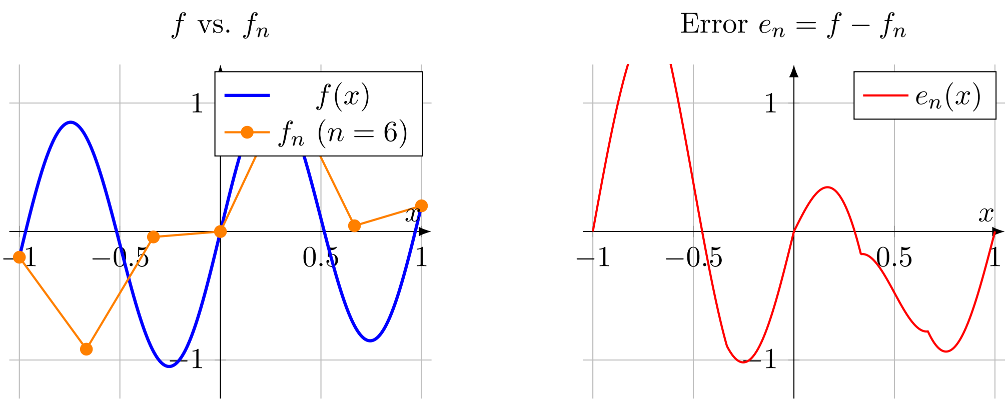


Figure 19: Left: function f and interpolant f_n ($n = 6$). Right: error curve $e_n = f - f_n$.

Exercise 6. Powers preserve uniform convergence (under a bound).powers-uniform Let $f_n : [0,1] \to \mathbb{R}$ converge uniformly to f. Suppose f is bounded. Show that for every positive integer m,

$$g_n(t) := f_n(t)^m \quad \text{converges uniformly on } [0,1] \text{ to} \quad g(t) := f(t)^m.$$

Definition 7. Uniform convergence. We say $f_n \to f$ uniformly on a set E if $\sup_{t\in E} |f_n(t) - f(t)| \to 0$ as $n \to \infty$.

Lemma 8. ***Power is Lipschitz on bounded intervals.*** *Fix $m \in \mathbb{N}$ and $M > 0$. For all $u, v \in [-M, M]$,*

$$|u^m - v^m| \leq m\, M^{m-1}\, |u - v|.$$

Proof. By the Mean Value Theorem applied to $\phi(x) = x^m$ on $[-M, M]$, $|u^m - v^m| = |\phi'(c)|\,|u - v| = m|c|^{m-1}|u-v| \le mM^{m-1}|u-v|$ for some c between u and v. Alternatively, use $u^m - v^m = (u-v)\sum_{k=0}^{m-1} u^{m-1-k}v^k$ and the bound $|u|, |v| \le M$. $\square$

Theorem 8. ***Uniform convergence is preserved by powers.*** *Let $f_n \to f$ uniformly on $[0,1]$ and suppose f is bounded. Then, for every $m \in \mathbb{N}$, $f_n^m \to f^m$ uniformly on $[0,1]$.*

Proof. Let $B := \sup_{t\in[0,1]} |f(t)| < \infty$. Since $f_n \to f$ uniformly, there exists N_0 such that $\sup_t |f_n(t) - f(t)| \le 1$ for all $n \ge N_0$. Hence for $n \ge N_0$, $|f_n(t)| \le B + 1 =: M$ for all t.

Fix $m \in \mathbb{N}$. By the lemma, for $n \ge N_0$ and all t,

$$|f_n(t)^m - f(t)^m| \le m\,M^{m-1}\,|f_n(t) - f(t)|.$$

Taking suprema over $t \in [0,1]$,

$$\sup_t |f_n(t)^m - f(t)^m| \le m\,M^{m-1} \sup_t |f_n(t) - f(t)| \xrightarrow[n\to\infty]{} 0,$$

so $f_n^m \to f^m$ uniformly. $\square$

Example 2. *Let $f(t) = \sin(2\pi t) + 0.2t$ and $f_n(t) = f(t) + \frac{1}{n}\cos(6\pi t)$ on $[0,1]$. Then* $\sup|f_n - f| \le 1/n$*, so $f_n \to f$ uniformly and f is bounded with* $\sup|f| \le 1.2$*. Taking $M = 2.2$, for $m = 3$,*

$$\sup|f_n^3 - f^3| \le 3M^2 \sup|f_n - f| \le \frac{14.52}{n} \to 0.$$

Thus $f_n^3 \to f^3$ uniformly; the same argument works for any m.

Remark 9. The boundedness of f is essential to produce a uniform Lipschitz constant for $x \mapsto x^m$ on the range of f_n and f. More generally, if $f_n \to f$ uniformly and ϕ is uniformly continuous on a bounded interval containing the ranges of f and all large n of f_n, then $\phi \circ f_n \to \phi \circ f$ uniformly.

Exercise 10. Runge approximation on a punctured domainrunge-annulus Let $B_r(z)$ be the open disc of radius r about $z \in \mathbb{C}$, and set $U := B_2(1) \setminus B_1(0)$. Suppose f is holomorphic on U.

(i) Show there exists a sequence converging to f uniformly on compact subsets of U.

(ii) Must there be a sequence of polynomials converging to f uniformly on U?

(iii) If, moreover, f is holomorphic on an open set containing $\overline{U}$, must there be a sequence of polynomials converging to f uniformly on U?

Definition 11. Laurent polynomial. A finite sum $\sum_{k=-N}^{M} a_k z^k$ with integers $N, M \ge 0$.

Theorem 9 (Runge, one–point–per–component form)**.** *If $K \subset \mathbb{C}$ is compact and $A \subset \widehat{\mathbb{C}} \setminus K$ contains at least one point from each component of $\widehat{\mathbb{C}} \setminus K$, then every function holomorphic on a neighborhood of K can be approximated uniformly on K by rational functions with poles only in A.*

Solution. (i) Fix a compact $K \subset U$. Then $\widehat{\mathbb{C}} \setminus K$ has two components: one contains ∞, the other contains 0. Apply Runge with $A = \{0, \infty\}$. The approximants are rational functions with poles in A, hence Laurent polynomials. They converge uniformly to f on K. Since K was arbitrary, the convergence holds on every compact subset of U.

(ii) No. Take $f(z) = 1/z$. If polynomials p_n converged uniformly to f on U, then for any simple closed curve $\gamma \subset U$ winding once around 0 we would have

$$\oint_\gamma p_n(z)\,dz \to \oint_\gamma \frac{1}{z}\,dz = 2\pi i,$$

but the left-hand side is 0 for all n (Cauchy's theorem), a contradiction.

(iii) Still no in general. The same $f(z) = 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, which contains $\overline{U}$, yet cannot be uniformly approximated on U by polynomials, by the argument in (ii). □

Remark 12. While polynomials fail in (ii)–(iii), Runge guarantees uniform approximation on U by *rational* (indeed, Laurent–polynomial) functions with poles at 0 (and possibly at ∞).

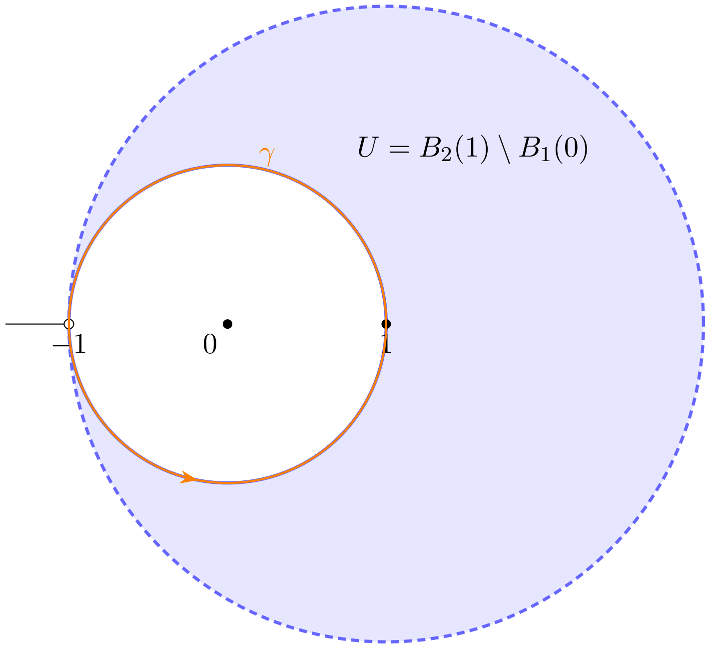


Figure 20: Domain U: open disc $B_2(1)$ with the open unit disc $B_1(0)$ removed. Dashed boundary is not in U; the inner circle is.

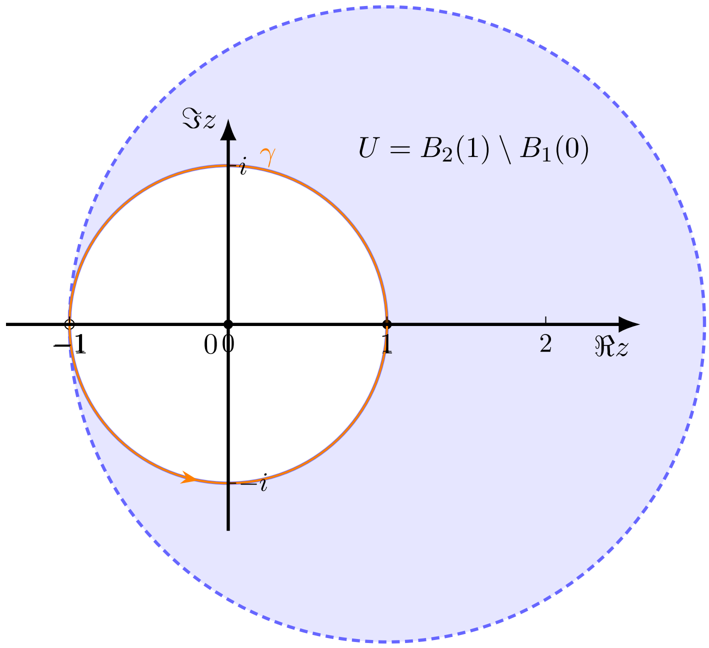


Figure 21: Domain U with axes and ticks. Dashed boundary is excluded; inner circle is included.

Example 3. ***Approximating*** $1/z$ ***on*** $U = B_2(1) \setminus B_1(0)$**.**

(i) *(Compacts in U.) Runge with poles at $\{0,\infty\}$ allows Laurent polynomials. For $f(z)=1/z$ the constant sequence*

$$r_n(z) \equiv \frac{1}{z}$$

already works and converges uniformly to $1/z$ on every compact $K \subset U$.

If $K \subset B_1(1)$ (so $|z-1|<1$ on K), the polynomials

$$p_N(z) = \sum_{k=0}^{N} (-1)^k (z-1)^k$$

satisfy $p_N \to 1/z$ uniformly on K by the geometric series $\frac{1}{1+w} = \sum_{k\geq 0}(-w)^k$ with $w = z-1$.

(ii) *(No polynomials on all of U.) Suppose p_n were polynomials with $p_n \to 1/z$ uniformly on U. For the loop $\gamma(t)=e^{it}$,*

$$\int_\gamma p_n(z)\,dz \longrightarrow \int_\gamma \frac{1}{z}\,dz = 2\pi i,$$

but each $\int_\gamma p_n(z)\,dz = 0$, contradiction. Hence no polynomial sequence converges to $1/z$ uniformly on U (and the same holds even if $1/z$ extends past $\overline{U}$).

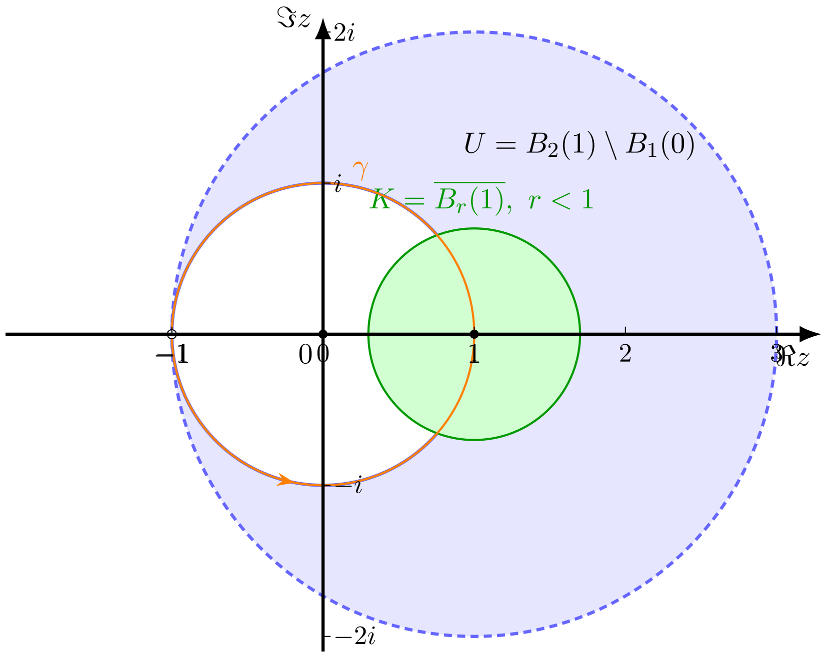


Figure 22: Domain U (blue), a compact $K \subset B_1(1)$ (green) where polynomials approximate $1/z$, and a loop γ around 0 used to rule out uniform polynomial approximation on all of U. Axes are extended for clarity.

Definition 13. Runge set / pole set. For a compact $K \subset \mathbb{C}$ and a set $A \subset \widehat{\mathbb{C}} \setminus K$, we say A is a *Runge pole set for K* if A meets every connected component of $\widehat{\mathbb{C}} \setminus K$.

Theorem 10. *Runge's theorem (rational form).* *Let $K \subset \mathbb{C}$ be compact and let $A \subset \widehat{\mathbb{C}} \setminus K$ meet every component of $\widehat{\mathbb{C}} \setminus K$. If f is holomorphic on a neighbourhood of K, then there exists a sequence of rational functions $\{r_n\}$ with all poles contained in A such that*

$$\sup_{z\in K} |r_n(z) - f(z)| \xrightarrow[n\to\infty]{} 0.$$

Corollary 11. ***Polynomial approximation (connected complement).*** *If $\widehat{\mathbb{C}}\setminus K$ is connected, then one may take $A = \{\infty\}$, hence f can be uniformly approximated on K by polynomials.*

Corollary 12. ***Laurent–polynomial approximation (one bounded hole).*** *If $\widehat{\mathbb{C}} \setminus K$ has exactly two components (the unbounded one and a bounded one containing a point a), then taking $A = \{\infty, a\}$ shows that f can be uniformly approximated on K by finite Laurent series in powers of $(z - a)$ (i.e. by sums $\sum_{k=-N}^{M} a_k(z - a)^k$).*

Remark 14. **Specialization to $U = B_2(1) \setminus B_1(0)$.** For any compact $K \subset U$, the complement $\widehat{\mathbb{C}} \setminus K$ has two components: one containing ∞, one containing 0. Runge with $A = \{\infty, 0\}$ yields uniform approximation on K by Laurent polynomials with possible pole at 0. In particular, for $f(z) = 1/z$, the constant sequence $r_n(z) \equiv 1/z$ (a rational function with pole at 0) already converges uniformly on every compact $K \subset U$. By contrast, no sequence of *polynomials* can converge uniformly to $1/z$ on all of U (use a loop $\gamma \subset U$ winding once around 0 and $\int_\gamma p(z)\, dz = 0$ versus $\int_\gamma \frac{1}{z}\, dz = 2\pi i$).

Exercise 15. Polynomial approximation of $1/z$ on a semicirclepoly-1-over-z-semi Construct a sequence of polynomials that converges uniformly to $1/z$ on

$$K = \{\, z \in \mathbb{C} : \ |z| = 1,\ \Re z \geq 0 \,\}.$$

Definition 16. Runge/Mergelyan (polynomial case). If $K \subset \mathbb{C}$ is compact with connected complement, every function continuous on K and holomorphic on $\operatorname{int} K$ can be uniformly approximated on K by polynomials.

Construction and proof of convergence. Set $t_k^{(n)} = -\frac{\pi}{2} + \frac{k\pi}{n}$ and $z_k^{(n)} = e^{it_k^{(n)}}$ for $k = 0, \dots, n$. Let

$$\ell_k^{(n)}(z) = \prod_{\substack{j=0\\ j\neq k}}^{n} \frac{z - z_j^{(n)}}{z_k^{(n)} - z_j^{(n)}}, \qquad p_n(z) = \sum_{k=0}^{n} \frac{1}{z_k^{(n)}}\, \ell_k^{(n)}(z).$$

Then p_n is a polynomial and $p_n(z_k^{(n)}) = 1/z_k^{(n)}$ for all k. Since K is an arc (its complement is connected) and $1/z$ is analytic on a neighbourhood of K, standard Bernstein–Walsh estimates for analytic data on Jordan arcs imply $p_n \to 1/z$ uniformly on K (in fact, $\|p_n - 1/z\|_K \leq C\rho^{-n}$ for some $\rho > 1$). This provides the required sequence. □

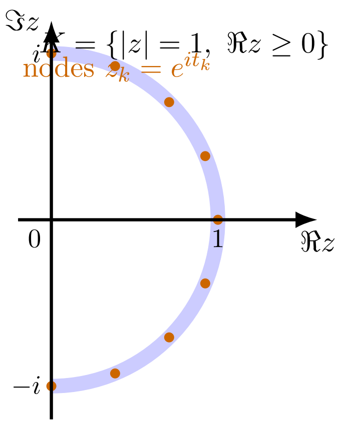
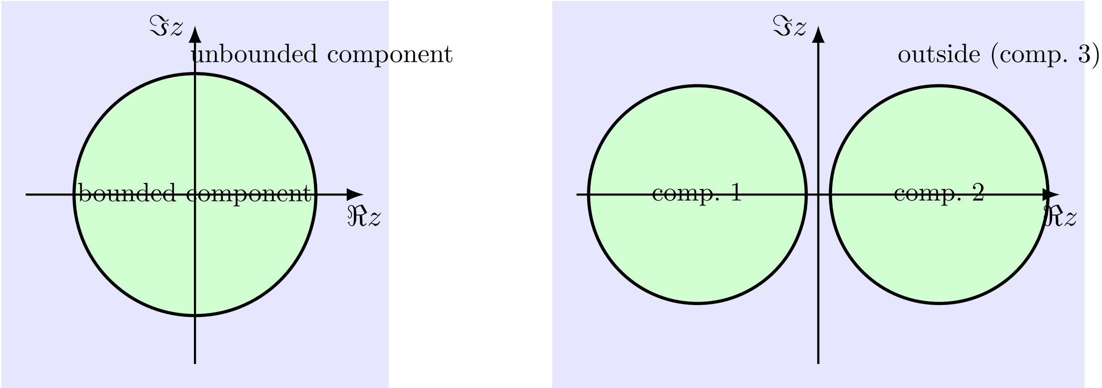


Figure 23: Right semicircle K and equally spaced interpolation nodes for the Lagrange polynomials p_n.



$K = \{|z| = 1\}$, $\mathbb{C} \setminus K$ has 2 components $\qquad$ $K = C_1 \cup C_2$, $\mathbb{C} \setminus K$ has 3 components

Figure 24: Complements of compact sets can be disconnected: a single circle separates the plane into two components; two disjoint circles give three components.

Example 4 (What the dots mean; explicit p_2)**.** *Let $K = \{|z| = 1,\ \Re z \geq 0\}$ and choose nodes $z_0 = -i$, $z_1 = 1$, $z_2 = i$. The Lagrange interpolant to $f(z) = 1/z$ is*

$$p_2(z) = \sum_{k=0}^{2} \frac{1}{z_k} \prod_{j \neq k} \frac{z - z_j}{z_k - z_j} = z^2 - z + 1.$$

Hence $p_2(1) = 1$, $p_2(i) = -i$, $p_2(-i) = i$, i.e. p_2 matches $1/z$ at the three orange nodes on the arc. For general n, pick $n+1$ nodes on K, form p_n by the same formula; then $p_n \to 1/z$ uniformly on K.

Exercise 17. Construct explicit polynomials $p_n \to 1/z$ on the right semicirclepoly-1-over-z-semicircle Let

$$K := \{z \in \mathbb{C} :\ |z| = 1,\ \Re z \geq 0\}.$$

Construct polynomials p_n such that $p_n \to 1/z$ uniformly on K.

Definition 18 (Lagrange interpolation)**.** Given $n+1$ distinct nodes $z_0, \ldots, z_n \in \mathbb{C}$, the Lagrange basis polynomials are

$$\ell_k(z) := \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{z - z_j}{z_k - z_j}, \qquad k = 0, \ldots, n,$$

so that $\ell_k(z_j) = \delta_{kj}$ and $\sum_{k=0}^{n} \ell_k \equiv 1$. For a function f defined on the nodes, the (unique) degree-$\leq n$ interpolant is

$$I_n[f](z) := \sum_{k=0}^{n} f(z_k)\, \ell_k(z).$$

Remark 19 (Barycentric evaluation (stable))**.** With *weights* $w_k := \left(\prod_{j \neq k}(z_k - z_j)\right)^{-1}$,

$$I_n[f](z) = \frac{\displaystyle\sum_{k=0}^{n} \frac{w_k\, f(z_k)}{z - z_k}}{\displaystyle\sum_{k=0}^{n} \frac{w_k}{z - z_k}},$$

which is numerically stable and avoids forming ℓ_k explicitly.

Definition 20 (Our nodes on the arc)**.** For each $n \in \mathbb{N}$, choose equally spaced angles

$$t_k^{(n)} := -\frac{\pi}{2} + \frac{k\pi}{n}, \qquad k = 0, \ldots, n,$$

and nodes on the right semicircle

$$z_k^{(n)} := e^{it_k^{(n)}} \in K.$$

(Other admissible choices are *Chebyshev-distributed* or *Leja* nodes on K.)

Theorem 13 (Construction)**.** *Let $f(z) = 1/z$. For $n \geq 1$ define the polynomial*

$$p_n(z) := \sum_{k=0}^{n} \frac{1}{z_k^{(n)}} \ell_k^{(n)}(z), \qquad \ell_k^{(n)}(z) := \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{z - z_j^{(n)}}{z_k^{(n)} - z_j^{(n)}}.$$

Then p_n is a polynomial of degree $\leq n$ and $p_n(z_k^{(n)}) = 1/z_k^{(n)}$ for all k.

Theorem 14 (Uniform convergence on K)**.** *With the above nodes, $p_n \to 1/z$ uniformly on K.*

Why the convergence holds (standard argument). The set K is a compact Jordan arc; its complement is connected. The function $1/z$ is holomorphic on a fixed annulus $\{1-\delta < |z| < 1+\delta\} \supset K$. By the *Bernstein–Walsh estimate*, the best approximation error

$$E_n := \inf_{\deg q \leq n} \| 1/z - q \|_K$$

decays geometrically: $E_n \leq C\rho^{-n}$ for some $\rho > 1$. For the Lagrange interpolant,

$$\| 1/z - p_n \|_K \;\leq\; (1 + \Lambda_n)\, E_n, \quad \Lambda_n := \sup_{z \in K} \sum_{k=0}^{n} \left|\ell_k^{(n)}(z)\right| \text{ (Lebesgue constant).}$$

For reasonable node families on smooth/analytic arcs (e.g. Chebyshev-type or Leja nodes), Λ_n grows at most polynomially in n, hence $(1 + \Lambda_n)E_n \to 0$ geometrically. Therefore $p_n \to 1/z$ uniformly on K. □

Example 5 (Make p_n concrete: $n = 2$)**.** *Take three nodes $z_0 = -i$, $z_1 = 1$, $z_2 = i$ on K (angles $-\frac{\pi}{2}, 0, \frac{\pi}{2}$). Then*

$$p_2(z) = \sum_{k=0}^{2} \frac{1}{z_k} \ell_k(z) = i\,\ell_0(z) + 1 \cdot \ell_1(z) - i\,\ell_2(z) \quad \Rightarrow \quad \boxed{p_2(z) = z^2 - z + 1}.$$

Indeed $p_2(1) = 1$, $p_2(i) = -i$, $p_2(-i) = i$ (matching $1/z$ at the nodes). On K, $\left|p_2(e^{it}) - e^{-it}\right| = \sqrt{\big(\cos 2t + 1 - 2\cos t\big)^2 + \big(\sin 2t\big)^2}$, so the uniform error already is modest and will decay rapidly as n grows.

Remark 21 (Geometric meaning). On the unit circle, $1/z = \overline{z}$. Thus we are approximating the *conjugation* map restricted to the right semicircle by polynomials in z. This is possible uniformly on the arc because its complement is connected (Mergelyan/Runge), even though conjugation is anti-holomorphic globally.

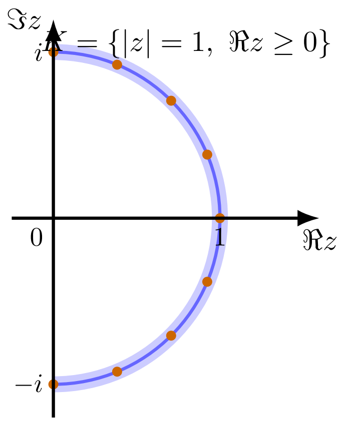


Figure 25: Right semicircle K and equally spaced interpolation nodes $z_k = e^{it_k}$.

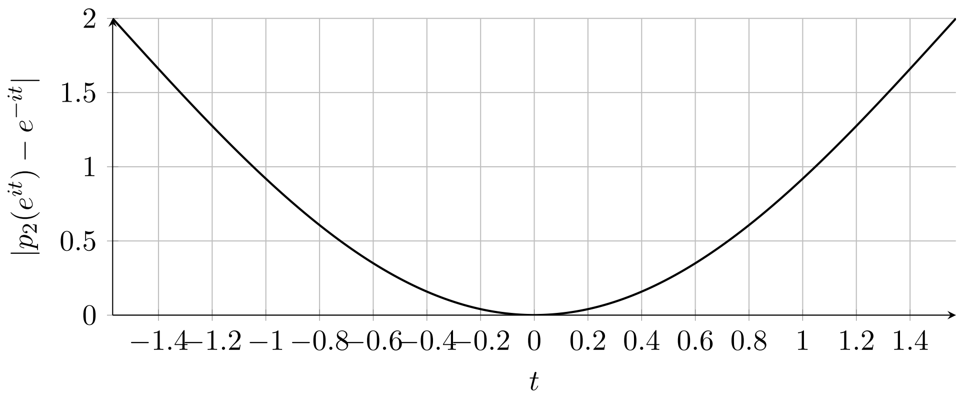


Figure 26: Pointwise error of $p_2(z) = z^2 - z + 1$ along $z = e^{it}$, $t \in [-\pi/2, \pi/2]$.

Exercise 22. Polynomial characterization of holomorphicity on Runge domainspoly-char Let $U \subset \mathbb{C}$ be bounded, open, and suppose $\mathbb{C} \setminus U$ is connected. Show that $f : U \to \mathbb{C}$ is holomorphic iff for every compact $K \Subset U$ and every $\varepsilon > 0$ there exists a polynomial P with

$$\sup_{z \in K} |f(z) - P(z)| < \varepsilon.$$

Definition 23. Polynomial hull in the plane. For a compact $K \subset \mathbb{C}$, let

$$\widehat{K} := K \cup \bigcup_{\Omega \text{ bounded component of } \mathbb{C} \setminus K} \overline{\Omega}.$$

Then $\mathbb{C} \setminus \widehat{K}$ is connected. If $K \Subset U$ and $\mathbb{C} \setminus U$ is connected, then $\widehat{K} \Subset U$.

Theorem 15 (Mergelyan)**.** *If $K \subset \mathbb{C}$ is compact with connected complement, and g is continuous on K and holomorphic on $\operatorname{int} K$, then for every $\varepsilon > 0$ there exists a polynomial P with $\sup_K |g - P| < \varepsilon$.*

Solution. ($\Rightarrow$) Assume f holomorphic on U, let $K \Subset U$, and set $\widehat{K}$ as above. Then $\widehat{K} \Subset U$ and $\mathbb{C} \setminus \widehat{K}$ is connected, hence by Mergelyan there is a polynomial P with $\sup_{\widehat{K}} |f - P| < \varepsilon$, thus in particular on K.

($\Leftarrow$) Assume: for every $K \Subset U$ and $\varepsilon > 0$ there is a polynomial P with $\sup_K |f - P| < \varepsilon$. Let $K_1 \Subset K_2 \Subset \cdots \Subset U$ with $\bigcup_m K_m = U$ and choose polynomials P_m with $\sup_{K_m} |f - P_m| < 1/m$.

Then $P_m \to f$ uniformly on each K_ℓ, so by Weierstrass' theorem the limit f is holomorphic on U. $\square$

Remark 24. On $U = \{\,|z| < 1\,\}$ one may take P_n to be partial sums of the Taylor series of f at 0, which converge uniformly on $K = \{\,|z| \le r\,\}$ for each $r < 1$. The theorem shows such polynomial approximation on every compact holds on any bounded U with connected complement.

Definition 25. Compact inclusion. For sets $K \subset U \subset \mathbb{C}$, we write $K \Subset U$ if $K \subset U$ and $\overline{K}$ is compact with $\overline{K} \subset U$. Equivalently, in the plane $\operatorname{dist}(K, \partial U) > 0$.

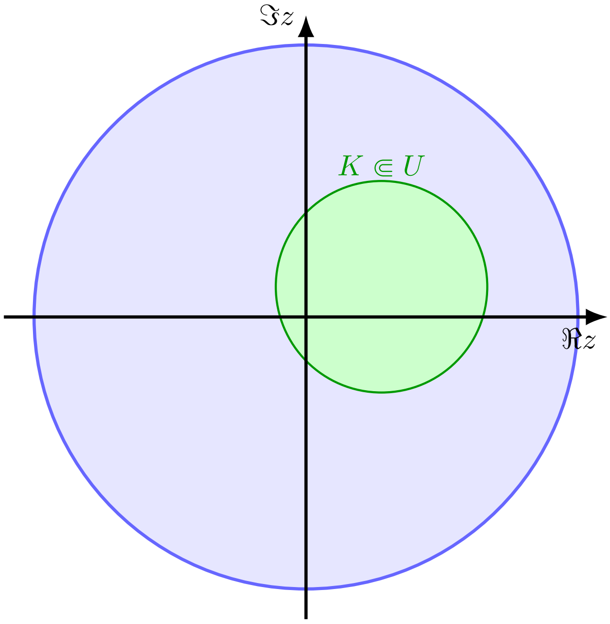


Figure 27: $K \Subset U$: K sits a positive distance away from ∂U.

Definition 26. Pointwise and uniform convergence. Let E be a set and (f_n) functions on E.

- $f_n \to f$ pointwise on E if $\forall x \in E$, $\forall \varepsilon > 0$, $\exists N = N(\varepsilon, x)$ with $|f_n(x) - f(x)| < \varepsilon$ for $n \ge N$.
- $f_n \to f$ uniformly on E if $\sup\limits_{x \in E} |f_n(x) - f(x)| \xrightarrow[n\to\infty]{} 0$.
- $f_n \to f$ *locally uniformly* on an open U if for every $K \Subset U$, the convergence is uniform on K.

Remark 27. On a compact set K, uniform on $K \iff$ convergence in the sup norm $\|\cdot\|_\infty$. If U is open, "locally uniform" means uniform on every $K \Subset U$; on a compact U this reduces to uniform on U.

Example 6 (Uniform vs nonuniform convergence)**.**

1. *Uniform:* $f_n(x) = \dfrac{x}{n+x}$ *on* $[0,1]$*; then* $\sup_{[0,1]} |f_n| = \frac{1}{n+1} \to 0$*, so* $f_n \to 0$ *uniformly.*
2. *Nonuniform:* $f_n(x) = x^n$ *on* $[0,1]$*; then* $f_n \to f$ *pointwise with* $f = \mathbf{1}_{\{1\}}$*, but not uniformly, since* $\sup_{x\in[0,1)} |x^n - 0| = 1$ *for all* n*.*

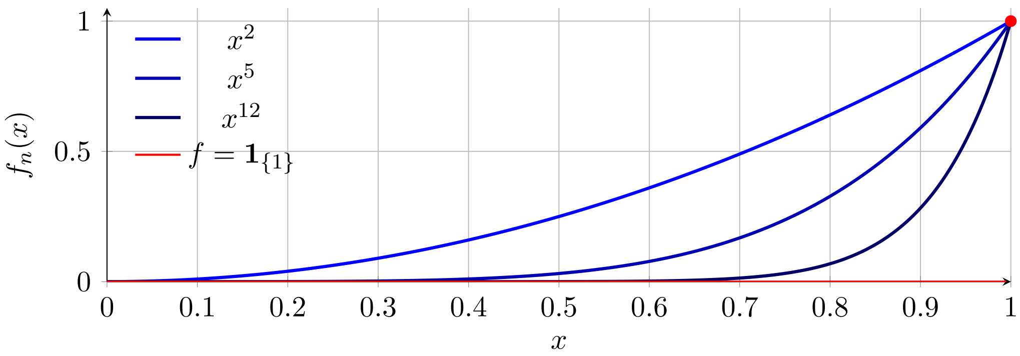


Figure 28: $x^n \to \mathbf{1}_{\{1\}}$ pointwise on $[0,1]$, but not uniformly.

Example 7 (Uniform vs nonuniform approximation by series)**.** *Let* $S_n(x) = \sum_{k=0}^{n} x^k$.

- *On* $[0,b]$ *with* $0 < b < 1$: $\left\| \frac{1}{1-x} - S_n \right\|_{[0,b]} = \sup_{[0,b]} \frac{x^{n+1}}{1-x} \le \frac{b^{n+1}}{1-b} \to 0$ *(uniform).*
- *On* $[0,1)$: *not uniform, since* $\sup_{x\in[0,1)} \dfrac{x^{n+1}}{1-x} = +\infty$.

Exercise 28. Polynomials p_n with $p_n(0) = 1$ and $p_n(z) \to 0$ for $z \neq 0$poly-vanish-off-zero Does there exist a sequence of complex polynomials (p_n) such that $p_n(0) = 1$ for all n and $p_n(z) \to 0$ for each $z \in \mathbb{C} \setminus \{0\}$?

Definition 29. For $n \in \mathbb{N}$ let

$$K_n := \left\{ z : \ \tfrac{1}{n} \le |z| \le n, \quad |\arg z| \le \pi - \tfrac{1}{n} \right\} \quad \text{(a “slit annulus”)}.$$

Then K_n is compact, $K_n \Subset \mathbb{C} \setminus \{0\}$, and $\bigcup_{n\ge1} K_n = \mathbb{C} \setminus \{0\}$. The complement $\mathbb{C} \setminus K_n$ is *connected* because the missing thin sector joins the inner and outer parts.

Theorem 16. *There exists a sequence of polynomials* (p_n) *with* $p_n(0) = 1$ *for all* n *and* $p_n(z) \to 0$ *for every* $z \neq 0$.

Proof. Fix n. Define a function on the compact set $E_n := K_n \cup \{0\}$ by

$$\phi_n(z) = \begin{cases} 0, & z \in K_n, \\ 1, & z = 0. \end{cases}$$

This ϕ_n is continuous on E_n and holomorphic on $\operatorname{int} E_n$ (the interior is the slit annulus, where $\phi_n \equiv 0$). Since $\mathbb{C} \setminus E_n = (\mathbb{C} \setminus K_n) \setminus \{0\}$ is still connected, Mergelyan's theorem applies: for each $\varepsilon_n > 0$ there exists a polynomial P_n with

$$\sup_{z\in E_n} |P_n(z) - \phi_n(z)| < \varepsilon_n.$$

Choose $\varepsilon_n \le \frac{1}{4}$ so that in particular $|P_n(0) - 1| < \frac{1}{4}$. Set

$$p_n(z) := \frac{P_n(z)}{P_n(0)}.$$

Then p_n is a polynomial with $p_n(0) = 1$, and for $z \in K_n$,

$$|p_n(z)| = \frac{|P_n(z)|}{|P_n(0)|} \le \frac{|P_n(z) - \phi_n(z)|}{|P_n(0)|} \le \frac{\varepsilon_n}{1 - \varepsilon_n} \le 2\varepsilon_n.$$

Now fix $z \neq 0$. For n large enough we have $z \in K_n$, hence $|p_n(z)| \le 2\varepsilon_n \to 0$ as $n \to \infty$. This proves the claim. $\square$

Remark 30. No uniformity on large discs is possible: if a polynomial q were tiny on the circle $|z| = R$, the maximum modulus principle on $|z| \le R$ would force $|q(0)|$ tiny as well. The slit ensures $\mathbb{C} \setminus K_n$ is connected, allowing Mergelyan, but K_n does *not* enclose 0, so there is no contradiction with $p_n(0) = 1$.

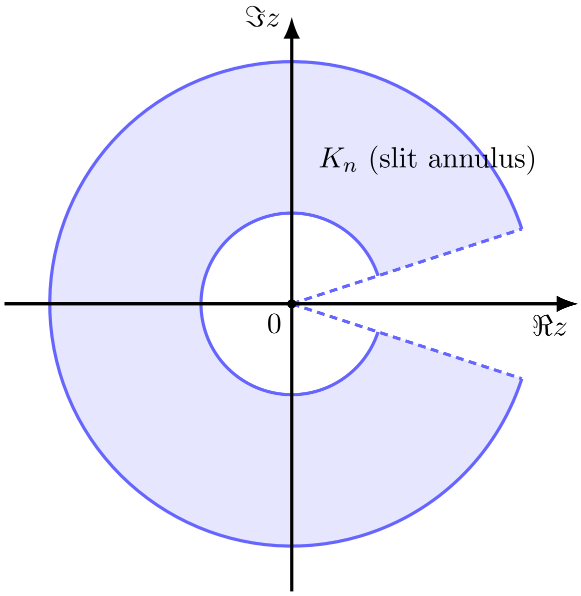


Figure 29: A typical K_n: annulus $\{1/n \le |z| \le n\}$ minus a thin sector; its complement is connected.

Exercise 31. Uniform polynomial approximation on an annulus forces extension -annulus-extension Let $A = \{z \in \mathbb{C} : \frac{1}{2} \le |z| \le 1\}$ and let $f : A \to \mathbb{C}$ be continuous on A and holomorphic on $\operatorname{int} A$. Suppose polynomials p_n converge uniformly to f on A. Show there exists $g : \{|z| \le 1\} \to \mathbb{C}$ continuous, holomorphic on $\{|z| < 1\}$, with $g = f$ on A.

Proof. Fix $\rho \in (\frac{1}{2}, 1)$. Since $|z| = \rho \subset A$ and $p_n \to f$ uniformly on A, the sequence (p_n) is uniformly Cauchy on $|z| = \rho$: $\sup_{|z|=\rho} |p_n - p_m| \to 0$ as $n, m \to \infty$. For each n, m the difference $p_n - p_m$ is entire, hence by the maximum modulus principle on the closed disk $\overline{D}_\rho := \{|z| \le \rho\}$,

$$\sup_{|z|\le\rho} |p_n(z) - p_m(z)| \le \sup_{|z|=\rho} |p_n(z) - p_m(z)| \xrightarrow[n,m\to\infty]{} 0.$$

Thus (p_n) converges uniformly on $\overline{D}_\rho$ to some function g_ρ that is holomorphic on $|z| < \rho$.

If $\rho' < \rho$ are both $> \frac{1}{2}$, then $p_n \to g_{\rho'}$ on $\overline{D}_{\rho'}$ and $p_n \to g_\rho$ on $\overline{D}_\rho$, so $g_{\rho'} = g_\rho$ on $|z| < \rho'$. Define g on $\{|z| < 1\} = \bigcup_{\rho > 1/2} \{|z| < \rho\}$ by $g := g_\rho$ on $|z| < \rho$; this is well defined and holomorphic.

On $A \cap \{|z| \le \rho\} = \{\frac{1}{2} \le |z| \le \rho\}$ we have $p_n \to f$ (by hypothesis) and $p_n \to g$ (by uniform convergence on $\overline{D}_\rho$), hence $g = f$ there. Letting $\rho \uparrow 1$ yields $g = f$ on all of A. Finally, g is continuous on $\{|z| \le 1\}$ because it is holomorphic on $\{|z| < 1\}$ and coincides with the continuous f on $|z| = 1$. $\square$

Remark 32. As a corollary, $1/z$ cannot be approximated uniformly by polynomials on A, since such an approximation would produce a holomorphic extension across 0.

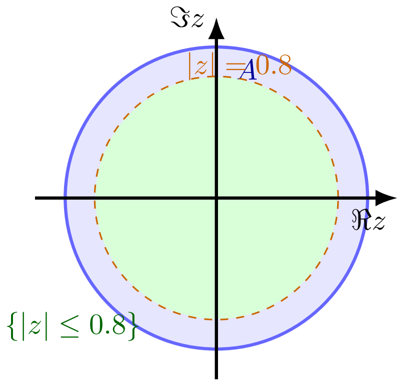


Figure 30: Fix $\rho \in (\frac{1}{2}, 1)$. Uniform convergence on $|z| = \rho$ and MMP make (p_n) Cauchy on $\{|z| \le \rho\}$, giving a holomorphic limit inside; letting $\rho \uparrow 1$ fills the disk and matches f on A.

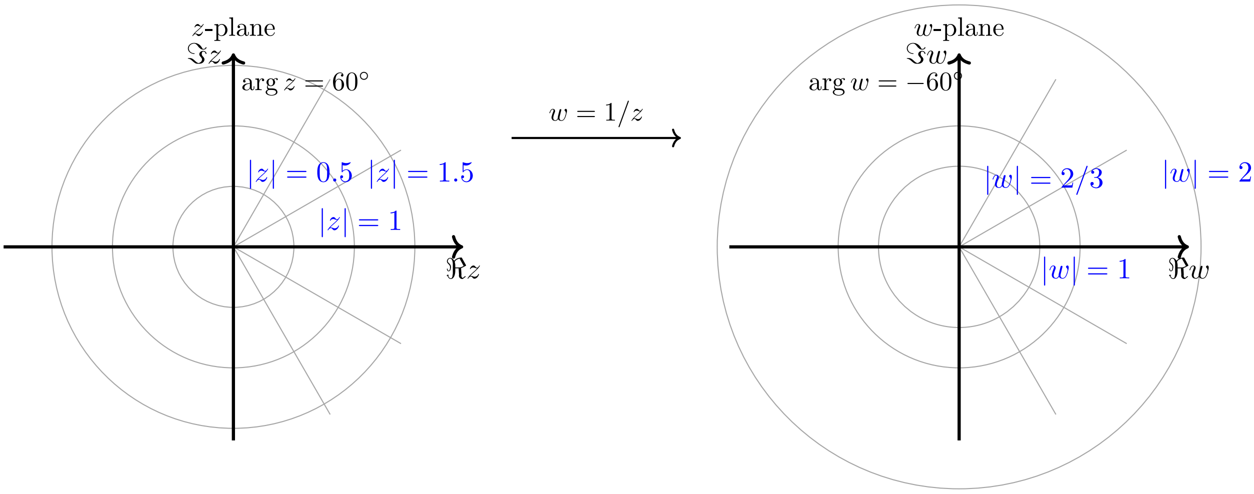


Figure 31: Circles centered at 0 invert radii, rays flip angles. The unit circle is fixed (parameters reverse).

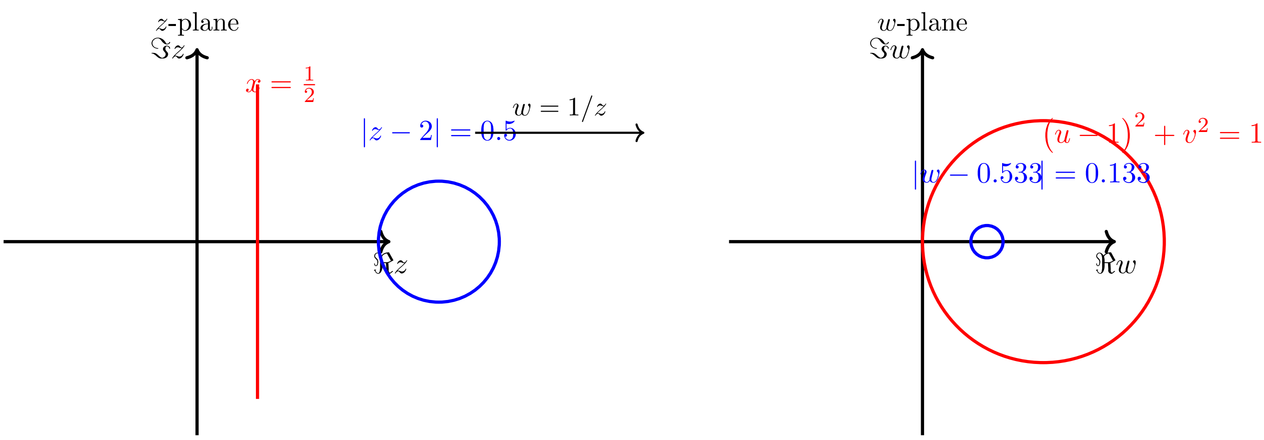


Figure 32: Under $w = 1/z$: lines not through 0 become circles through 0; circles not through 0 remain circles with transformed center/radius.

Exercise 33. Irrationality and algebraicityirr-alg

1. **Irrational sums/powers.**
   (a) $(\sqrt{3}+\sqrt{5}) \notin \mathbb{Q}$. If $\sqrt{3}+\sqrt{5} = r \in \mathbb{Q}$, then $r^2 = 8+2\sqrt{15}$, hence $\sqrt{15} = (r^2-8)/2 \in \mathbb{Q}$, impossible.

   (b) $e^2 \notin \mathbb{Q}$. Suppose $e^2 = a/b \in \mathbb{Q}$. For $N \geq \max\{2, b\}$, both $N!e^2$ and $N!\sum_{n=0}^{N} \frac{2^n}{n!}$ are integers, but

   $$0 < N!\Big(e^2 - \sum_{n=0}^{N} \frac{2^n}{n!}\Big) = \sum_{n=N+1}^{\infty} \frac{2^n N!}{n!} < \sum_{k=1}^{\infty}\Big(\frac{2}{N+1}\Big)^k < 1,$$

   a contradiction.

2. **Algebraic numbers and coefficients.**
   (a) A real α is algebraic iff it is a zero of some nonzero polynomial with rational coefficients. ($\Leftarrow$) Clear after clearing denominators. ($\Rightarrow$) Integers are rationals.

   (b) *False* that rational coefficients are forced. Example: $(x-\sqrt{2})(x-\sqrt{3}) = x^2 - (\sqrt{2}+\sqrt{3})x + \sqrt{6}$. The roots are algebraic, but the coefficients are not rational.